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A Theoretical Reduction From Membership Inference to Reconstruction Attacks Against Learners

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Abstract

We study *reconstruction attacks*, which recover training data from a learned model. In our adversarial model, the adversary designs a hard instance, drawn from a parameterized family of data distributions, and is given the parameter used to generate it. Our main contribution is a general reduction showing that, in this model, *membership inference* (determining whether a given datapoint was used for training) implies reconstruction: for a training set of size n, a membership test with recall at least $q(n)$ and false-positive rate $O(\delta^2/n^2)$ yields a reconstruction algorithm that, with probability $1 - O(\delta)$, outputs only training points, and recovers $\Omega(n \cdot q(n))$ of them in expectation. This reduction applies to any learning algorithm. For stochastic convex optimization (SCO), we make the reduction *computationally efficient* over exponentially large domains via structured distributions whose support is partitioned into polynomially many categories of exponentially many elements each, so that reconstruction iterates over categories rather than the full support. Combining this with an existing membership inference result for SCO, we prove that any learning algorithm achieving low excess risk over all distributions admits a polynomial-time reconstruction attack recovering $\Omega(n)$ training points, even when the underlying distribution has exponentially large support.

1 INTRODUCTION

Despite their widespread adoption, machine learning models remain vulnerable to a range of information retrieval attacks, exploits that leak private information through model parameters. These threats vary in what the adversary aims to learn:

- *Property inference*: inferring global statistical properties of the training data that the model producer did not intend to share, such as demographic composition or environmental conditions (Ateniese et al., 2015; Ganju et al., 2018).
- *Membership inference*: determining whether a specific datapoint was part of the training data (Shokri et al., 2017; Yeom et al., 2018).
- *Model inversion*: exploiting model access to infer sensitive training data features, ranging from recovering representative inputs for a class to inferring unknown attributes of known individuals (Fredrikson et al., 2014, 2015).
- *Reconstruction*: directly extracting actual training data from the model. Such attacks can expose sensitive personal information by, for example, extracting verbatim text including names, phone numbers, and email addresses from a language model, or reconstructing training images from neural network classifiers (Carlini et al., 2019, 2021; Balle et al., 2022; Haim et al., 2022).

The ability to extract training data implies that models "memorize" portions of their training data. One might hope to mitigate this risk by avoiding memorization entirely. However, Feldman (2020) established that for long-tailed distributions, which are characteristic of natural data, memorization of individual labels, including noisy and outlier labels, is *necessary* for achieving close-to-optimal generalization error. This creates a fundamental tension: the very property that enables learning may also enable information retrieval attacks.

Differential privacy (DP) offers one approach to this problem. DP schemes provide mathematically proven guarantees on information leakage by adding noise during training (Dwork et al., 2006; Abadi et al., 2016).

In principle, DP can reduce leaked information to arbitrarily small levels, making it the primary formal defense against the privacy vulnerabilities described above. However, achieving meaningful DP guarantees comes at a significant cost to model utility: the noise required for strong privacy typically degrades accuracy (Carlini et al., 2021), and the assumptions underlying DP – that private information maps neatly to individual training records – may not hold for natural language and other complex data (Brown et al., 2022).

This raises a fundamental question: differential privacy greatly restricts information leakage, but is such a strong guarantee necessary to prevent reconstruction?

To address this question rigorously, we begin by examining the relationship between membership inference and reconstruction. Membership inference determines whether a *given* candidate belongs to the training set, while reconstruction must *output* training data, knowing at most their distribution. Reconstruction is a stronger requirement: a reconstruction algorithm implicitly solves membership inference for every datapoint it outputs, but the converse fails in general; we show that it holds when the adversary designs the data distribution (Section 2).

Membership inference has received substantial theoretical attention. Theoretical bounds quantitatively relate the success of membership inference to overfitting (Yeom et al., 2018), and theoretically optimal attack strategies have been derived for various settings (Sablayrolles et al., 2019; Carlini et al., 2022). By contrast, reconstruction attacks have been studied primarily empirically – demonstrating extraction of training data from language models (Carlini et al., 2021) – or theoretically only for restricted model classes (Haim et al., 2022; Balle et al., 2022; Voitovych et al., 2025; Cohen et al., 2024). General theoretical bounds for reconstruction, however, have been largely absent.

This gap between membership inference theory and reconstruction practice motivates our central question: *Can membership inference guarantees be systematically converted into reconstruction guarantees?* If so, the rich theoretical toolkit for analyzing membership inference could be leveraged to establish reconstruction bounds.

We study the bridging of this gap in both a general context and in the *stochastic convex optimization* (SCO) setting. SCO provides an ideal setting for our investigation for two reasons. First, it underlies gradient-based optimization, the workhorse behind modern machine learning (Shalev-Shwartz et al., 2009). Second, the gap between private and non-private learning is sharp and well-quantified: without privacy constraints, $\Theta(1/\varepsilon^2)$ samples suffice for learning (Agarwal et al., 2012; Feldman, 2016), while DP-compliant algorithms require sample complexity that grows with the dimension d (Bassily et al., 2014), making SCO a natural setting to ask whether the full strength of DP is necessary to prevent reconstruction, or whether the privacy cost can be avoided.

Formally, in SCO a learner receives n datapoints from a distribution $\mathcal{D}$ over a domain $\mathcal{Z}$ and outputs a parameter $\hat{\theta}$ minimizing the population risk $F(\theta) = \mathbb{E}_{z\sim\mathcal{D}}[f(\theta, z)]$, where both $f(\cdot, z)$ and Θ are convex. The goal is to achieve excess risk $F(\hat{\theta}) - \min_\theta F(\theta) \leq \varepsilon$ with minimal sample complexity n.

In the privacy field under this setting, recent work by Attias et al. (2024) established an important impossibility result. Any *ε-learner*, an algorithm that achieves excess risk at most ε for *all* distributions (i.e., with distribution-free guarantees), must necessarily "memorize" its training data in a precise sense. Specifically, for any ε-learner there exists a (learner-dependent) hard distribution for which a membership inference test correctly identifies training datapoints with constant probability while maintaining a low false-positive rate. This result advances the membership inference research in SCO, and our approach aims to translate this advancement, as well as similar ones, to reconstruction.

The existence of a fundamental setting – stochastic convex optimization – where a membership test is inevitably highly probable raises two questions: (i) *Is reconstruction also always highly probable in this setting?* (ii) *Can it be done efficiently?* Our aim is to answer these two questions and provide a framework for answering them for future settings as well.

Our contributions. We take a step towards bridging the gap between membership inference and reconstruction with three main results, establishing a general theoretical framework for reconstruction attacks on learning algorithms.

1. Reduction from Membership Inference to Reconstruction. Our first contribution is framework-agnostic, and does not require the assumptions made in SCO. We show that if a setting admits a per-learner hard distribution for membership inference, then the same setting admits a per-learner hard distribution for reconstruction: the reconstruction-hard distribution is uniform over a candidate set sampled from the inference-hard distribution, provided the test keeps its guarantees under it (Definition 2.3). Quantitatively, for a training set of size n, if the data distribution is uniform over a candidate set and a membership test has recall at least $q(n)$ (Definition 2.3) and false-positive rate $O(\delta^2/n^2)$, then an algorithm that applies the test to every candidate outputs, with probability $1 - O(\delta)$, only training datapoints, and $\Omega(n \cdot q(n))$ of them in

expectation. This reduction is *learner-agnostic* and *test-agnostic*: it applies to any learner and any membership test satisfying the stated conditions.

2. Efficient Reconstruction on Exponential Domains. *A general construction.* The reduction in (1) enumerates over the candidate set, so its runtime scales with the size of the data domain, which is prohibitive when the domain is exponentially large. Our second contribution addresses this limitation: we give a construction that makes reconstruction computationally efficient even over exponentially large supports, by modeling the distribution as a polynomially large set of categories, each of which is an exponentially large set. We argue that this structure mirrors a feature found in many natural data distributions: a polynomial number of high-level categories, each populated by exponentially many fine-grained instances. For example, the set of all natural images is exponentially large, but it can be split into a (still large, but not exponential) set of trigram descriptions, e.g., "three golden triangles", "plumber eating spaghetti", "swirly purple fruit", etc.

We construct a distribution that fits this model, for which the adversary produces a single candidate from each category set; with probability $1 - O(\delta)$, for a constant fraction of the training points, the candidate of the corresponding category coincides exactly with that training point. Such an adversary thus only needs a sufficiently differentiating inference test while iterating over the polynomial set of categories.

3. Impossibility Theorem for SCO. *An application to SCO.* Combining the above, we establish a strong impossibility result: any ε-learner (with distribution-free guarantees) is inherently vulnerable to reconstruction attacks. We obtain this by instantiating the category construction of (2) inside the SCO setting and establishing, for our own membership test and under the constructed block-uniform distribution, the two guarantees the reduction consumes: recall, carried over to it from the correlation bound of Attias et al. (2024, Lemma 6.3) (Appendix C.5), and soundness (a low false-positive rate), which we prove directly, since their product-measure tail analysis does not apply to it (Lemma 3.3, Corollary 3.4). The approach is modular: the category construction is setting-agnostic, and we instantiate it in SCO because that is, to our knowledge, the only learning setting with a membership inference impossibility at the rates the reduction requires; any future such result plugs into our reduction once its two rates hold under the hard distribution of Definition 2.2.

Together, these results establish that for the linear SCO instance in the optimal sample complexity regime, any ε-learner is vulnerable to reconstruction of $\Omega(n)$ training datapoints on some distribution of our family. Thus distribution-free accuracy precludes reconstruction resistance across distributions, a tension that goes beyond differential privacy.

Organization. Section 2 presents our reduction; Section 3 develops the exponential-domain distribution construction and proves the main SCO reconstruction result; Section 4 surveys related work. Full proofs and supporting lemmas appear in Appendices A–D (notation is summarized in Appendix D), and Appendix E discusses generalizations, necessary features of the adversarial model, and open directions.

2 REDUCTION FROM MEMBERSHIP INFERENCE TO RECONSTRUCTION

In this section, we establish our first main result: a general reduction converting membership inference capabilities into reconstruction attacks. This reduction is learner-agnostic and applies to any setting with a membership test that is sufficiently accurate under the hard distribution of Definition 2.2.

2.1 Problem Setup

We consider a general learning setting where a *learning algorithm* $\mathcal{A} = \{\mathcal{A}_n\}_{n\in\mathbb{N}}$ is given as a sequence of maps $\mathcal{A}_n : \mathcal{Z}^n \to \Theta$, one for every sample size: on a training set $S_n = \{z_1, \dots, z_n\}$ of datapoints drawn from a distribution over a domain $\mathcal{Z}$, it outputs a model $\hat{\theta} = \mathcal{A}_n(S_n) \in \Theta$ (uppercase Z_i denotes a random datapoint, lowercase z a realized one). In stochastic convex optimization (Section 3) the learning problem is a *convex problem instance*: a triple $(\Theta, \mathcal{Z}, f)$ with Θ a convex subset of Euclidean space and the loss $f(\cdot, z)$ convex for every $z \in \mathcal{Z}$; the reduction of this section requires no convexity.

Definition 2.1 (Membership inference test)**.** *A membership inference test is a function $T : \Theta \times \Theta \times \mathcal{Z} \to \{0, 1\}$ that attempts to determine whether a datapoint $x \in \mathcal{Z}$ was part of the training set, given the learned parameter $\hat{\theta}$, the parameter θ of the data-generating process, and the datapoint x.*

Definition 2.2 (Adversarial reconstruction game)**.** *Fix a sample size n, a candidate-set size N, and a confidence parameter $\delta > 0$.*

1. *The adversary designs a parameterized family of distributions $\{\mathcal{D}_\theta\}_{\theta\in\Theta}$ over the domain $\mathcal{Z}$ and a prior $\mathbf{D}$ over the parameters $\theta \in \Theta$.*
2. *Nature draws $\theta \sim \mathbf{D}$. The adversary, knowing θ, samples a candidate set $\Sigma = \{\sigma_1, \dots, \sigma_N\} \sim \mathcal{D}_\theta^{\otimes N}$ and defines the* hard distribution $\Delta_\theta := \mathrm{Uniform}(\Sigma)$.

3. *A training set $S \sim \Delta_\theta^{\otimes n}$ is drawn; the learner receives S only, and outputs $\hat{\theta} = \mathcal{A}_n(S)$.*
4. *The adversary receives $(\hat{\theta}, \theta)$ and applies the membership test T to the candidates in Σ.*

The adversary thus constructs the family, the prior, and the hard distribution (steps 1–2) and is handed θ (step 4), so it has nothing to estimate. Some per-learner choice of hard distribution is *necessary* at the standard sample size (Appendix E.2), while exact knowledge of θ relaxes to sampling access at polynomial cost in the SCO instantiation (Appendix E.1).

Theorem 2.4 below assumes two guarantees on the membership test, defined next; both are required under the constructed Δ_θ, the distribution the learner is actually trained on, not under the ambient $\mathcal{D}_\theta$. Section 3 establishes both for the SCO instance under its own hard distribution Δ_μ: soundness in Section 3.2, and recall through the correlation bound (Lemma B.2, proven for the product Bernoulli family $\mathcal{D}_\mu$ and applied to Δ_μ in Appendix C.5).

Definition 2.3 (Recall and false-positive rates)**.** *Let T be a membership inference test, played in the game of Definition 2.2. T has* recall *at least $q(n)$ and* false-positive rate *at most β, with confidence δ, if with probability $1-\delta$ over $\theta \sim \mathbf{D}$ and the adversary's construction of Δ_θ (the sampling of Σ), the following hold, where all inner probabilities and expectations are over $S \sim \Delta_\theta^{\otimes n}$ and any internal randomness of $\mathcal{A}$ and T:*

$$\mathbb{E}\Big[\big|\{x \in S : T(\mathcal{A}_n(S), \theta, x) = 1\}\big|\Big] \geq q(n) \cdot n, \quad (1)$$

$$\mathbb{P}\big[T(\mathcal{A}_n(S), \theta, x) = 1 \mid x \notin S\big] \leq \beta \quad \text{for every candidate } x \in \Sigma. \quad (2)$$

The conditioning in (2) is well defined, as each candidate lands in S with probability $\approx n/N$ under the finite-support Δ_θ. Condition (1) says that the test flags, in expectation, at least a $q(n)$ fraction of the training points (its true-positive rate over the training set); this count form is what Step 5 of the proof of Theorem 2.4 uses.

2.2 Main Reduction Theorem

Theorem 2.4 (Reduction from Membership Inference to Reconstruction)**.** *Consider the game of Definition 2.2 with $N = n^2/\delta$, played by a learning algorithm $\mathcal{A} = \{\mathcal{A}_n\}_{n\in\mathbb{N}}$, a family $\{\mathcal{D}_\theta\}$, and a prior $\mathbf{D}$. Suppose that, under the game's hard distribution Δ_θ, the membership inference test T has recall at least $q(n)$ and false-positive rate $O(\delta^2/n^2)$ with confidence δ (Definition 2.3), and that $\delta = o(q(n))$. Then there is a reconstruction algorithm (Algorithm 1) that, with probability $1 - O(\delta)$, returns $S' \subseteq S$ with $\mathbb{E}[|S'|] = \Omega(n \cdot q(n))$.*

Algorithm 1 Basic Reconstruction

```
Input: θ̂, θ, candidate set Σ = {σ_1, ..., σ_N}
Output: Reconstructed set S'
S' ← ∅
for i = 1, ..., N do
    if T(θ̂, θ, σ_i) = 1 then
        S' ← S' ∪ {σ_i}
return S'
```

2.3 Proof of Theorem 2.4

Proof overview. The key idea is to discretize the domain into a finite candidate set Σ, small enough to search exhaustively but large enough to avoid collisions among training samples (Steps 1–2). The membership test is then applied to every candidate: its low false-positive rate ensures that only true training points are flagged (Step 3), while its recall guarantees that many are recovered (Steps 4–5).

Step 1: Distribution construction. Steps 2–3 of the game (Definition 2.2) are played with $N := |\Sigma| = n^2/\delta$: the adversary samples $\Sigma = \{\sigma_1, \ldots, \sigma_N\} \sim \mathcal{D}_\theta^{\otimes N}$ from the ambient family and defines the *discrete uniform distribution* $\Delta_\theta := \mathrm{Uniform}(\Sigma)$, from which the training sample $S \sim \Delta_\theta^{\otimes n}$ is drawn; the conditions of Definition 2.3 are required under Δ_θ.

Step 2: Collision bound. When $N \geq n^2/\delta$, i.i.d. sampling from Δ_θ rarely produces collisions, so with high probability each training datapoint corresponds to a distinct element of Σ. Specifically, $\mathbb{P}[\text{each } \sigma_i \text{ sampled at most once}] = \prod_{k=0}^{n-1} \frac{N-k}{N} \geq (1 - \frac{n}{N})^n \geq (1 - \frac{\delta}{n})^n \geq 1 - \delta$, where the last inequality is Bernoulli's inequality $(1-x)^n \geq 1 - nx$ for $x \in [0, 1]$. Therefore, with probability at least $1-\delta$, all elements of S are distinct, so the flagged training points counted by condition (1) correspond to as many distinct recovered candidates.

Step 3: No false positives. Writing $T(\sigma)$ for $T(\hat{\theta}, \theta, \sigma)$, the union bound over the N candidates and condition (2) give

$$\begin{aligned}\mathbb{P}[\exists \sigma \in \Sigma : T(\sigma) = 1 \wedge \sigma \notin S] &\leq N \cdot \max_{\sigma \in \Sigma} \mathbb{P}[T(\sigma) = 1 \mid \sigma \notin S] \\ &\leq O\left(N \cdot \frac{\delta^2}{n^2}\right) = O(\delta).\end{aligned}$$

Thus, with probability $1 - O(\delta)$, the test has no false positives within the candidate set Σ, ensuring $S' \subseteq S$.

Step 4: Reconstruction algorithm. Algorithm 1 implements the reconstruction by exhaustively testing all candidates in Σ.

Step 5: Expected reconstruction size. Let E denote the intersection of the no-collision event (Step 2, probability $\geq 1-\delta$) and the no-false-positive event (Step 3, probability $\geq 1-O(\delta)$), so $\mathbb{P}[E] \geq 1-O(\delta)$. Let $X = |\{x \in S : T(x) = 1\}|$ be the number of flagged training points; the recall condition (1) gives $\mathbb{E}[X] \geq q(n)\cdot n$. On the event E the flagged training points are distinct and are all returned, so $|S'| \geq X - n\cdot \mathbb{1}[E^c]$ pointwise (trivially on E^c). Taking expectations,

$$\begin{aligned}\mathbb{E}[|S'|] \;&\geq\; \mathbb{E}[X] - n\cdot\mathbb{P}[E^c] \;\geq\; q(n)\cdot n - O(\delta)\cdot n \\ &=\; (1-o(1))\,q(n)\cdot n \;=\; \Omega(n\cdot q(n)),\end{aligned}$$

where the first equality uses the hypothesis $\delta = o(q(n))$. This completes the proof.

3 RECONSTRUCTION ON EXPONENTIAL DOMAINS

The reduction in Section 2 tests a candidate set Σ exhaustively, so a polynomial runtime needs a polynomial-size set, whereas natural distributions have exponentially large support (2^d for d-dimensional binary data). In this section we overcome this barrier with distributions partitioned into polynomially many sets, each containing exponentially many elements; we call these sets "categories", as the members of each share a common property.

We specialize to SCO, with the convex problem instance (Section 2) fixed and explicit: $\Theta = \{\theta \in \mathbb{R}^d : \|\theta\| \leq 1\}$, $\mathcal{Z} = \{\pm 1/\sqrt{d}\}^d$, and the linear (hence convex) loss $f(\theta, z) = -\langle\theta, z\rangle$, i.e. mean estimation, the standard hard instance of the SCO lower-bound literature (Attias et al., 2024). The parameter θ of Section 2 becomes the distribution mean μ, $\mathcal{D}_\theta$ becomes the product Bernoulli family $\mathcal{D}_\mu$, and the learner outputs $\hat{\theta} \in \Theta$; the role of the hard distribution Δ_θ of Definition 2.2 is played by the block-uniform Δ_μ of Section 3.1, under which both guarantees of Definition 2.3 are established.

Definition 3.1 (Block-uniform distribution)**.** *A distribution Δ over a domain $\mathcal{Z}$ is* block-uniform *with N categories if its support can be partitioned into N disjoint* categories $C_1, \ldots, C_N$*, and Δ samples by drawing $l \sim$ Uniform$([N])$ and then $z \sim$ Uniform(C_l).*

In our construction, $N = \mathrm{poly}(n)$ while $|C_l| = 2^{d^{1/4}} - 2 \geq 2^{d^{1/4}-1}$ for every l, so the support is exponentially large but organized into polynomially many categories.

Theorem 3.2 (Impossibility over Exponential Domains)**.** *Let $(\Theta, \mathcal{Z}, f)$ be the linear SCO instance above, and let $\mathcal{D}_\mu$ be the product distribution over $\mathcal{Z}$ with $\mathbb{P}[z_i = +1/\sqrt{d}] = \frac{1}{2}(1+\mu_i\sqrt{d})$ independently for each coordinate, where $\mu \sim$ Uniform$([-12\varepsilon/\sqrt{d},\ 12\varepsilon/\sqrt{d}]^d)$ so that $\mathbb{E}[z_i] = \mu_i$.*

There exists $\varepsilon_0 = 1/(12\sqrt{2})$ such that for every $\varepsilon \leq \varepsilon_0$ and $\delta \leq \varepsilon$ with $\delta^{-1} \in$ poly(n), every n in the regime $n = \Theta(1/\varepsilon^2)$, and every $d \geq d_0 = \Omega\big(\max\{n^2\log(n+\delta^{-1}),\ \varepsilon^4 n^4/\delta^2,\ n^8\log^4(n/\delta)\}\big)$, the following holds: for every ε-learner $\mathcal{A}_n$ there exists a block-uniform distribution Δ_μ, drawn from the explicitly constructed family of Section 3.1, whose support consists of $N\cdot(2^{d^{1/4}}-2)$ vectors ($N = n^2/\delta$ categories of $2^{d^{1/4}}-2$ vectors each), such that, for a training set $S \sim \Delta_\mu^{\otimes n}$, with probability $\geq 1-O(\delta)$, Algorithm 2 reconstructs $\Omega(n)$ points of S in poly(n, d, δ^{-1}) time.

Here $n = \Theta(1/\varepsilon^2)$ abbreviates the window $\Omega(1/\varepsilon^2) \leq n \leq 1/(13824\,\varepsilon^2)$ (Appendix D explains both ends). The quantifier order matters: the hard distribution is chosen *per learner*, and some such dependence is necessary at the standard sample size (Appendix E.2), whereas Algorithm 2 is a *single* explicit algorithm working against every ε-learner. Sections 3.1–3.3 together constitute the proof.

3.1 Proof Overview and Construction

Proof overview. The exhaustive search of Theorem 2.4 is infeasible over the exponentially large domain $\mathcal{Z}$. We overcome this by partitioning the support into polynomially many *categories*, each with a rigid internal structure: some coordinates are pinned to match the distribution mean, while the rest are block-constant. This structure makes each category decodable, so given the learned model, we can recover the unique training point in a category without enumerating it. The proof combines four ingredients: (1) the correlation bound (Lemma B.2, proven for product samples and applied to Δ_μ in Appendix C.5) guarantees that most training points leave a detectable signal in $\hat{\theta}$; (2) the category tail bound (Lemma 3.3) ensures that non-training points do not produce false positives; (3) sign consistency (Lemma B.4) guarantees that the decoder recovers each flagged training point exactly; and (4) the algorithm runs in polynomial time by iterating over the polynomially many categories.

The construction has two layers, fixed by the following partition structure. Each category $l \in [N]$ is governed by a partition of the coordinates

$$\Pi_l \;=\; \big(\pi_{l,0},\ \pi_{l,1},\ \ldots,\ \pi_{l,d^{1/4}}\big),$$

consisting of the *mean-alignment part* $\pi_{l,0}$, whose coordinates are *pinned* to fixed values chosen (by the bit-fitting rule below) so that the mean of the resulting distribution matches μ, hence the name, and the $d^{1/4}$ *blocks* $\pi_{l,1}, \ldots, \pi_{l,d^{1/4}}$, on which every member of category l is constant. The intuition: pinned coordinates make $\mathbb{E}[z] \approx \mu$; blocks make categories decodable.

Partition sampling. For each category $l \in [N]$ ($N = n^2/\delta$), construct $\Pi_l = (\pi_{l,0}, \pi_{l,1}, \ldots, \pi_{l,d^{1/4}})$: include each $i \in [d]$ in $\pi_{l,0}$ with probability $\varepsilon/\sqrt{d}$ independently; assign remaining coordinates uniformly to $d^{1/4}$ bins (Algorithm 3 in Appendix A).

Bit-fitting. For each $i \in [d]$, let $K_i = \{l : i \in \pi_{l,0}\}$. Choose pin values $(B_{i,l})_{l \in K_i} \in \{\pm 1\}^{|K_i|}$ to minimize $|\mu_i - \frac{1}{N}\sum_{l \in K_i} B_{i,l}/\sqrt{d}|$ (Algorithm 4 in Appendix A).

Category definition. Category C_l consists of all $z \in \{\pm 1/\sqrt{d}\}^d$ satisfying: (i) $z_i = B_{i,l}/\sqrt{d}$ for $i \in \pi_{l,0}$ (pinned), and (ii) $z_i = z_{i'}$ for all $i, i' \in \pi_{l,k}$, $k \in [d^{1/4}]$ (block-constant), excluding the two vectors whose $d^{1/4}$ block signs all coincide (all blocks positive, or all negative), so $|C_l| = 2^{d^{1/4}} - 2$. To sample $x \sim \Delta_\mu$: draw $l \sim \mathrm{Uniform}([N])$, then $x \sim \mathrm{Uniform}(C_l)$. The categories are pairwise disjoint except with probability $e^{-\Omega(\varepsilon\sqrt{d})}$ over the partitions (Lemma B.8); disjointness can also be enforced deterministically by trimming, leaving Algorithm 2 unchanged (Remark B.9).

Mean-matching. For coordinate i: if $i \notin \pi_{l,0}$, the block sign is symmetric (the excluded pair is sign-flip symmetric) so $\mathbb{E}[x_i \mid l] = 0$; if $i \in \pi_{l,0}$, then $x_i = B_{i,l}/\sqrt{d}$. Averaging over l: $\mathbb{E}[x_i] = \frac{1}{N}\sum_{l \in K_i} B_{i,l}/\sqrt{d} \approx \mu_i$ by bit-fitting.

3.2 Soundness of the Membership Test under Δ_μ

This subsection verifies the false-positive condition (2) of Definition 2.3 for the constructed Δ_μ, where the technical difficulty of this section is concentrated: Δ_μ is *not* a product distribution (coordinates within a block are perfectly correlated), so the product-measure sub-Gaussian soundness argument of Attias et al. (2024) does not apply. Lemma 3.3 and Corollary 3.4 restore it; without them, the argument of Section 2 could not be carried over to Δ_μ.

Lemma 3.3 (Category tail bound)**.** *Fix category l, draw $z \sim$ Uniform(C_l), and let $M_l = \max_k |\pi_{l,k}|$. For fixed $y \in \mathbb{R}^d$,*

$$\mathbb{P}[\langle y, z - \mathbb{E}[z \mid l]\rangle \geq t] \leq 2\exp\left(-\frac{t^2 d}{2M_l\|y\|^2}\right).$$

Proof sketch. Adding back the two excluded vectors makes the block signs independent Rademachers, so Hoeffding's inequality applies (Cauchy–Schwarz within each block yields the factor M_l); removing this sign-flip symmetric pair leaves $\mathbb{E}[z \mid l]$ unchanged and at most doubles the tail (full proof in Appendix C.1). □

Corollary 3.4 (Soundness under Δ_μ)**.** *Let $A = \mathrm{diag}(g(\varepsilon, \mu))$ be the reweighting matrix of Attias et al. (2024, Lemma 6.3) (restated as Definition B.1). Setting $y = A\hat{\theta}$ with $\|A\|_2 \leq 288\varepsilon^2$ (Lemma B.3) and $\|\hat{\theta}\| \leq 1$, assuming $n = \Theta(1/\varepsilon^2)$ as in Theorem 3.2 (in particular $n \leq 1/(13824\,\varepsilon^2)$) and $d = \Omega(M \cdot n^2 \log(n/\delta))$ where $M = \max_l M_l$, this ensures $\mathbb{P}\Big[\exists i \in [n] : \langle A\hat{\theta}, Z_i - \mu\rangle \geq \varepsilon/(2n)\Big] \leq O(\delta^2/n^2)$, for fresh $Z_1, \ldots, Z_n \sim \Delta_\mu$ (using the center-shift bound, Lemma C.1); the slack below the test's threshold ε/n is consumed in Appendix E.1.*

Algorithm 2 Partition-Based Reconstruction (with subroutines Algorithms 3–4, App. A)

Input: $\hat{\theta}$; μ; dimension d; $g(\varepsilon, \mu)$ (Def. B.1); partitions $(\Pi_l)_{l=1}^N$ and pin values $(B_{i,l})$, built by the adversary via Algorithms 3–4 (App. A)
Output: Reconstructed set S'
$S' \leftarrow \emptyset$
for $l = 1, \ldots, N$ **do**
  $\forall i \in \pi_{l,0}$: $z'_i \leftarrow B_{i,l}/\sqrt{d}$
  for $k = 1, \ldots, d^{1/4}$ **do**
    $\forall i \in \pi_{l,k}$:
      $z'_i \leftarrow \mathrm{sgn}\Big(\sum_{j \in \pi_{l,k}} g(\varepsilon, \mu)_j\, \hat{\theta}_j\Big)/\sqrt{d}$
  if $\langle g(\varepsilon, \mu) \odot (z' - \mu),\ \hat{\theta}\rangle \geq \varepsilon/n$ **then**
    $S' \leftarrow S' \cup \{z'\}$
return S'

Remark D.1 traces the three terms of d_0 to mean alignment, sign consistency, and soundness, and resolves the self-referential requirement on d in Corollary 3.4, where the maximum block size M itself depends on d.

3.3 The Reconstruction Algorithm

For each category l, Algorithm 2 constructs a candidate z' by: (i) setting pinned coordinates to their pin values, and (ii) decoding block-constant coordinates via the sign of the reweighted block score $\sum_{j \in \pi_{l,k}} g(\varepsilon, \mu)_j \hat{\theta}_j$, where $g(\varepsilon, \mu)_i$ is the coordinate-wise reweighting scalar defined in Definition B.1, and $\odot$ denotes the Hadamard (entrywise) product. It then applies the membership test, here the thresholding rule $T(\hat{\theta}, \mu, x) = \mathbb{1}[\langle g(\varepsilon, \mu) \odot (x - \mu),\ \hat{\theta}\rangle \geq \varepsilon/n]$, which instantiates the abstract test $T : \Theta \times \Theta \times \mathcal{Z} \to \{0, 1\}$ of Definition 2.1.

Appendix B analyzes these ingredients, Appendix C gives the deferred proofs (the full sign-consistency derivation is Appendix C.3), and Appendix C.5 assembles the complete proof.

4 RELATED WORK

Reconstruction attacks and traceability. Reconstruction from aggregate data is not a new topic: in an early result, Dinur and Nissim (2003) showed that

answering too many linear queries on a database enables reconstruction of nearly the entire dataset – a result that motivated the development of differential privacy; see Dwork et al. (2017) for a broad survey of reconstruction and re-identification attacks. In the learning setting, Carlini et al. (2019) demonstrated unintended memorization in neural networks, and Carlini et al. (2021) extracted verbatim training data from large language models; Nasr et al. (2023) later scaled such extraction to production language models, while Carlini et al. (2023a) extracted training images from diffusion models. On the theoretical side, Balle et al. (2022) analyzed reconstruction by informed adversaries with partial knowledge of the training set (e.g., all but one point), and Haim et al. (2022) developed reconstruction algorithms for neural network classifiers. Cohen et al. (2024) proposed formal definitions and protection notions for reconstruction attacks, identifying when an attack should count as convincing evidence of vulnerability, and Guo et al. (2022) provided upper bounds on reconstruction success under DP, subsequently tightened by Hayes et al. (2023) for DP-SGD.

Closest to our setting, Voitovych et al. (2025) studied traceability in ℓ_p SCO, showing that ε-learners are vulnerable to traceability attacks. Traceability – identifying which individuals contributed to a dataset given aggregate statistics – was formalized by Dwork et al. (2015), building on the genomic re-identification attacks of Homer et al. (2008). The fingerprinting code technique underlying many traceability lower bounds originated with Tardos (2008) and was adapted to differential privacy by Bun et al. (2014) and Steinke and Ullman (2017). Our work goes beyond traceability and membership inference to establish full reconstruction: recovering the actual training datapoints themselves.

Membership inference. Shokri et al. (2017) introduced membership inference attacks against machine learning models, using shadow models to distinguish training set members from non-members. Yeom et al. (2018) connected membership inference to the generalization gap, and Nasr et al. (2019) extended these attacks to the white-box and federated settings. Sablayrolles et al. (2019) derived Bayes-optimal membership inference strategies, and Carlini et al. (2022) developed a principled likelihood-ratio framework characterizing when and why attacks succeed. Choquette-Choo et al. (2021) further showed that membership inference remains effective even when the adversary observes only hard label predictions. Our reduction (Theorem 2.4) converts any membership test that is sufficiently accurate under the hard distribution of Definition 2.2, however constructed, into a reconstruction attack.

Memorization and generalization. Classical stability-based analyses by Bousquet and Elisseeff (2002) established generalization guarantees for algorithms whose output changes little under single-sample perturbations, a notion closely tied to the memorization–leakage trade-off at the heart of our work. In SCO, Shalev-Shwartz et al. (2010) showed that learnability can hold even in the absence of uniform convergence, highlighting the distinctive role of algorithmic stability. Zhang et al. (2017) demonstrated that deep networks can fit random labels, challenging capacity-based explanations of generalization, and Arpit et al. (2017) gave a complementary empirical characterization of memorization dynamics during training. Feldman (2020) proved that memorization of individual examples is *necessary* for near-optimal generalization on long-tailed distributions, and Feldman and Zhang (2020) validated this empirically via influence estimation. Brown et al. (2021) characterized when memorization of "irrelevant" training data is unavoidable for high-accuracy learning. Carlini et al. (2023b) measured how memorization scales with model size across language models.

Differential privacy and private SCO. Differential privacy (Dwork et al., 2006; Dwork and Roth, 2014) provides a rigorous framework for limiting information leakage from computations on private data. Kasiviswanathan et al. (2011) characterized what can be learned under differential privacy in the PAC model, and Abadi et al. (2016) introduced DP-SGD, the standard practical tool for training neural networks under differential privacy. In the SCO setting, Chaudhuri et al. (2011) initiated the study of differentially private empirical risk minimization, and Bassily et al. (2014) established tight dimension-dependent excess-risk bounds for private convex ERM. Jain and Thakurta (2014) showed that near dimension-independent rates are attainable under additional structural assumptions. Bassily et al. (2019) achieved optimal rates for private SCO, showing that the optimal excess population loss is the larger of the non-private SCO rate and the private-ERM rate, and Feldman et al. (2020) gave the first linear-time optimal algorithm. Asi et al. (2021) extended these results to ℓ_1 geometry. Our results complement these bounds: every ε-learner trained at the non-private sample size $n = \Theta(1/\varepsilon^2)$ (with $d \geq d_0$) is vulnerable to reconstruction (Theorem 3.2).

Information complexity in SCO. SCO was studied by Shalev-Shwartz et al. (2009), who established its learnability structure and the failure of uniform convergence and ERM in the general case, while Agarwal et al. (2012) established information-theoretic oracle-complexity lower bounds. Feldman (2016) showed that

ERM requires dimension-dependent sample complexity, and Livni (2024) recently analyzed the sample complexity of gradient descent itself in SCO. Stability-based analyses of generalization – pioneered by Hardt et al. (2016) for SGD – provide a complementary perspective, with Bassily et al. (2016) establishing algorithmic stability as a unifying lens on adaptive data analysis.

Information-theoretic generalization bounds using mutual information were developed by Russo and Zou (2020) and Xu and Raginsky (2017), and refined through the conditional mutual information framework of Steinke and Zakynthinou (2020). Haghifam et al. (2023) identified limitations of these bounds for gradient descent in SCO, and Livni (2023) proved information-theoretic lower bounds on the tightness of such bounds.

Most relevant to our work, Attias et al. (2024) combined these threads, proving that ε-learners in SCO must memorize their training data in an information-theoretic sense, with the learned model necessarily revealing large conditional mutual information about individual training points. We build directly on this result and on their correlation bound (their Lemma 6.3): we establish the membership test's recall and soundness under a non-product block-uniform distribution, and show that such memorization enables not just membership inference but full reconstruction, even over exponentially large supports.

5 CONCLUSION

We established a general theoretical framework for reconstruction attacks on learning algorithms, with three main contributions. First, we presented a learner-agnostic reduction (Theorem 2.4) showing that any membership inference test with low false-positive rate and non-negligible recall under a hard distribution of polynomial support enables efficient reconstruction of training points drawn from it. Second, we developed a block-uniform distribution construction (Theorem 3.2) that enables polynomial-time reconstruction over exponentially large domains, using partition structure to reduce the search space from 2^d to $\mathrm{poly}(n, d, \delta^{-1})$. Third, importing the correlation bound of Attias et al. (2024, Lemma 6.3) and establishing our membership test's recall and soundness under the constructed block-uniform distribution itself, we showed that any ε-learner with distribution-free guarantees is inherently vulnerable to reconstruction of $\Omega(n)$ training datapoints.

These results complete a chain of impossibility: Attias et al. (2024) proved that achieving low excess risk (ε-learning) requires memorization of training data in an information-theoretic sense, and that this memorization enables membership inference. Our work shows that membership inference, in turn, enables reconstruction. Together, these results establish that *in SCO, no learner can achieve low excess risk on all distributions while preventing reconstruction on all of them*; the obstacle lies in the learning problem, not in a particular algorithm.

These are worst-case statements: they do not say that a particular model trained on particular data is vulnerable, but that every ε-learner with distribution-free guarantees is vulnerable on some distribution in the family.

Limitations. The adversary chooses the hard distribution per learner and is given its parameter; the former is necessary at the standard sample size (Appendix E.2), the latter relaxes to sampling access (Appendix E.1). Our analysis assumes the SCO setting with convex losses and the product Bernoulli family as the ambient family (the hard distribution itself is block-uniform); extensions to non-convex settings or more general distribution families remain open. The dimension requirement $d \geq \Omega(\max\{n^2 \log(n + \delta^{-1}),\ \varepsilon^4 n^4/\delta^2,\ n^8 \log^4(n/\delta)\})$, whose third, soundness-driven term dominates for $\delta^{-1} = O(n^3)$ (Remark D.1), is large in practice and grows with the sample size, and Theorem 3.2 operates in the regime $n = \Theta(1/\varepsilon^2)$. Our results are asymptotic, and the constants hidden in $\Omega(\cdot)$ and $O(\cdot)$ notation matter for practical attacks.

Future directions. Several directions merit further investigation: (1) the conditions under which reconstructions are preventable; (2) the effects of adaptive defenses and their fundamental limitations; and (3) technical improvements to our framework, including tighter dimension bounds that reduce the polynomial dependence on n, formal characterization of when natural data distributions exhibit block-uniform structure, and precise lower bounds on the privacy–accuracy trade-off. The extension we consider most promising, to arbitrary exponential families via the generalized fingerprinting lemma of Kamath et al. (2022), is stated as an open problem in Appendix E.5; Appendix E also treats continuous distributions (Appendix E.4) and the category axioms (Appendix E.3).

AI use statement

In this work, we used generative AI tools for assisting in the writing of proofs and for assisting with translation. We have not used generative AI tools for helping develop theoretical models or conceptual frameworks, formulating mathematical claims, providing critical ingredients for proving mathematical claims, or proposing or refining hypotheses; generating synthetic data sets, designing or providing feedback on research methodology or experiments, implementing methods, cleaning

and reformatting datasets, supporting qualitative and thematic data analysis, and interpreting results are not applicable to this work. Additionally, we used generative AI tools for searching for relevant literature, editing phrases to improve readability, and formatting references. We have reviewed all AI-assisted work. All AI-assisted proof text was checked line by line by the authors. We take responsibility for the final content of this work, including text, claims, or artifacts produced with the aid of generative AI.

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A Theoretical Reduction From Membership Inference to Reconstruction Attacks Against Learners: Supplementary Materials

A ADDITIONAL ALGORITHMS

The main text presents two reconstruction algorithms: Algorithm 1 (Section 2) for polynomial-sized candidate sets, and Algorithm 2 (Section 3) for exponential domains. Below are the partition and distribution sampling subroutines used by Algorithm 2.

A.1 Partition Sampling

Algorithm 3 Partition Sampling

Input: Dimension d, number of categories N, parameter ε
Output: Partitions $\Pi_1, \ldots, \Pi_N$
for $l = 1, \ldots, N$ **do**
  $\pi_{l,0} \leftarrow \emptyset$
  for $i = 1, \ldots, d$ **do**
    if $\mathrm{Bern}(\varepsilon/\sqrt{d}) = 1$ **then**
      $\pi_{l,0} \leftarrow \pi_{l,0} \cup \{i\}$
  Initialize $\pi_{l,1}, \ldots, \pi_{l,d^{1/4}} \leftarrow \emptyset$
  for $i \in [d] \setminus \pi_{l,0}$ **do**
    $\mathrm{bin} \sim \mathrm{Uniform}(\{1, \ldots, d^{1/4}\})$; $\pi_{l,\mathrm{bin}} \leftarrow \pi_{l,\mathrm{bin}} \cup \{i\}$
  $\Pi_l \leftarrow (\pi_{l,0}, \pi_{l,1}, \ldots, \pi_{l,d^{1/4}})$
return $\{\Pi_1, \ldots, \Pi_N\}$

A.2 Distribution Sampling

Algorithm 4 Distribution Sampling (Bit-Fitting + Category Definition)

Input: Mean $\mu \in \mathbb{R}^d$, partitions $\{\Pi_l\}_{l=1}^N$
Output: Pin values $(B_{i,l})$, category sets $\{C_1, \ldots, C_N\}$
for $i = 1, \ldots, d$ **do**
  $K_i \leftarrow \{l \in [N] : i \in \pi_{l,0}\}$
  Choose $(B_{i,l})_{l \in K_i} \in \{\pm 1\}^{|K_i|}$ minimizing $|\mu_i - \frac{1}{N} \sum_{l \in K_i} B_{i,l}/\sqrt{d}|$
for $l = 1, \ldots, N$ **do**
  $C_l \leftarrow \{z \in \{\pm 1/\sqrt{d}\}^d : (\forall i \in \pi_{l,0},\ z_i = B_{i,l}/\sqrt{d})$
    $\wedge\ (\forall k \in [d^{1/4}],\ \forall i, i' \in \pi_{l,k},\ z_i = z_{i'})$
    $\wedge$ not all block signs of z coincide$\}$
return $(B_{i,l})$, $\{C_1, \ldots, C_N\}$

Complexity of Algorithm 2: $O(N \cdot d) = O(n^2 d/\delta)$, polynomial when $d = \mathrm{poly}(n, 1/\delta)$.

B TECHNICAL ANALYSIS

This section establishes the key technical ingredients for Theorem 3.2: the correlation bound, the reweighting vector, and sign consistency of block scores. Full proofs are deferred to Appendix C.

B.1 Reweighting Vector and Correlation

Definition B.1 (Reweighting vector). *For $\varepsilon > 0$ and $\mu \in [-12\varepsilon/\sqrt{d}, 12\varepsilon/\sqrt{d}]^d$, define*

$$g(\varepsilon,\mu)_i = \frac{144\varepsilon^2 - \mu_i^2 d}{1-\mu_i^2 d}, \qquad v(z)_i = g(\varepsilon,\mu)_i(z_i-\mu_i).$$

Lemma B.2 (Correlation bound (Attias et al., 2024)). *Under the uniform prior $U = Uniform([-12\varepsilon/\sqrt{d}, 12\varepsilon/\sqrt{d}]^d)$, for any ε-learner $\mathcal{A}_n$, $\mathbb{E}_{\mu\sim U, S_n\sim\mathcal{D}_\mu^{\otimes n}}\left[\sum_{j=1}^n \langle v(z_j), \hat{\theta}\rangle\right] \geq 6\varepsilon - 4\delta$. In particular, with probability $\geq 1-\delta$, at least $\Omega(n)$ training datapoints satisfy $\langle v(z_j), \hat{\theta}\rangle \geq \varepsilon/n$.*

Lemma B.3 (Reweighting norm bound). *For $\varepsilon \leq \varepsilon_0 = 1/(12\sqrt{2})$ and any $J \subseteq [d]$: $|g(\varepsilon,\mu)_i| \leq 288\varepsilon^2$ and $\sum_{i\in J} g(\varepsilon,\mu)_i^2 \leq 288^2\varepsilon^4|J|$.*

Proof. See Appendix C. □

B.2 Sign Consistency

The reconstruction algorithm sets block k's value to $\text{sign}(S_{l,k})/\sqrt{d}$ where $S_{l,k} = \sum_{i\in\pi_{l,k}} g(\varepsilon,\mu)_i\hat{\theta}_i$. For a datapoint $z \in C_l$, the *block sign* $s_k \in \{-1,+1\}$ is the common value $z_i\sqrt{d}$ for all $i \in \pi_{l,k}$ (which is well-defined since z is block-constant within category C_l). For correctness, we need $\text{sign}(S_{l,k}) = s_k$ (the training datapoint's block sign) for all k.

Lemma B.4 (Sign consistency). *Suppose $d \geq \Omega(\varepsilon^4 n^4/\delta^2)$ and $n \leq 1/(13824\,\varepsilon^2)$. For a training datapoint z from category C_l satisfying $\langle v(z), \hat{\theta}\rangle \geq \varepsilon/n$, we have sign$(S_{l,k}) = s_k$ for all $k \in [d^{1/4}]$ simultaneously, with probability $\geq 1 - O(\delta)$ over the random partition.*

Proof sketch. The correlation bound decomposes by blocks as $\frac{1}{\sqrt{d}}\sum_{k=1}^{d^{1/4}} s_k S_{l,k} \geq \frac{\varepsilon}{n} - |S_0| - |\mu\text{-term}| \geq \frac{\varepsilon}{2n}$, where S_0 is the contribution of the pinned coordinates and the μ-term $\sum_i g(\varepsilon,\mu)_i\mu_i\hat{\theta}_i$ arises from splitting $z_i - \mu_i = s_k/\sqrt{d} - \mu_i$ in non-pinned blocks. The μ-term is at most $3456\varepsilon^3$ by Cauchy–Schwarz (Lemma B.3), which is at most $\varepsilon/(4n)$ exactly when $n \leq 1/(13824\,\varepsilon^2)$; the pinned contribution satisfies $|S_0| \leq O(\varepsilon^{5/2}d^{-3/16})$ (Lemma B.6), below $\varepsilon/(4n)$ at the theorem's dimension.

Thus the average signed block score is $\geq \varepsilon d^{1/4}/(2n)$. Each block score has variance $\leq 288^2\varepsilon^4 d^{-1/4}$ over the random partition (Lemma B.5). By Chebyshev with $\kappa = \sqrt{d^{1/4}/\delta}$, the per-block deviation is $\leq 288\varepsilon^2/\sqrt{\delta}$, dimension-free. This is smaller than the signal $\varepsilon d^{1/4}/(2n)$ exactly when $d^{1/4} \geq 576\varepsilon n/\sqrt{\delta}$, i.e., $d \geq \Omega(\varepsilon^4 n^4/\delta^2)$. A union bound over all $d^{1/4}$ blocks completes the argument. The full proof appears in Appendix C.3. □

B.3 Supporting Lemmas

Lemma B.5 (Variance of block scores). *For any block $k \in [d^{1/4}]$, $Var(S_{l,k}) \leq 288^2\varepsilon^4 d^{-1/4}$.*

Proof. Each coordinate i is assigned to block k independently with probability $d^{-1/4}$. The variance is $\sum_i g(\varepsilon,\mu)_i^2\hat{\theta}_i^2 d^{-1/4}(1-d^{-1/4}) \leq d^{-1/4}\cdot\|g(\varepsilon,\mu)\|_\infty^2\|\hat{\theta}\|^2 \leq d^{-1/4}\cdot 288^2\varepsilon^4$, using $\|\hat{\theta}\| \leq 1$ and Lemma B.3. (The per-coordinate estimate $\hat{\theta}_i^2 \leq 1$ would give only $288^2\varepsilon^4 d^{3/4}$, which grows with d; the norm bound is what makes the Step 3 comparison in Appendix C.3 close.) □

Lemma B.6 (Bounded contribution of $\pi_{l,0}$). *For a fixed category l, with probability $1-\delta$ over the random partition, $|\pi_{l,0}| \leq 2\varepsilon d^{5/8}/\sqrt{\delta}$ and the pinned contribution $S_0 = \sum_{i\in\pi_{l,0}} g(\varepsilon,\mu)_i\hat{\theta}_i(z_i-\mu_i)$ satisfies*

$$|S_0| \;\leq\; 288\varepsilon^2\cdot\frac{2}{\sqrt{d}}\cdot\sqrt{|\pi_{l,0}|} \;\leq\; 576\sqrt{2}\,\varepsilon^{5/2}\,\delta^{-1/4}\,d^{-3/16} \;=\; O(\varepsilon^{5/2}d^{-3/16}),$$

which decays with d.

Proof. $\mathbb{E}[|\pi_{l,0}|] = \varepsilon\sqrt{d}$. Since each coordinate is included independently with probability $\varepsilon/\sqrt{d}$, the variance is $\mathrm{Var}(|\pi_{l,0}|) \le \varepsilon\sqrt{d}$. By Chebyshev's inequality,

$$\mathbb{P}\Big[|\pi_{l,0}| > \varepsilon\sqrt{d} + t\Big] \le \frac{\varepsilon\sqrt{d}}{t^2}.$$

Setting $t = \varepsilon\sqrt{d}\cdot d^{1/8}/\sqrt{\delta}$ (so $\varepsilon\sqrt{d}/t^2 = \delta/(\varepsilon\sqrt{d}\cdot d^{1/4}) \le \delta$ for d large) gives $|\pi_{l,0}| \le \varepsilon\sqrt{d} + \varepsilon d^{5/8}/\sqrt{\delta} \le 2\varepsilon d^{5/8}/\sqrt{\delta}$ with probability $1-\delta$. For S_0, each summand satisfies $|g(\varepsilon,\mu)_i| \le 288\varepsilon^2$ (Lemma B.3) and $|z_i - \mu_i| \le 2/\sqrt{d}$, so by Cauchy–Schwarz over $\pi_{l,0}$, using $\|\hat\theta\| \le 1$,

$$|S_0| \;\le\; 288\varepsilon^2 \cdot \frac{2}{\sqrt{d}} \sum_{i\in\pi_{l,0}} |\hat\theta_i| \;\le\; 288\varepsilon^2 \cdot \frac{2}{\sqrt{d}} \cdot \sqrt{|\pi_{l,0}|}\,\|\hat\theta\| \;\le\; 576\sqrt{2}\,\varepsilon^{5/2}\,\delta^{-1/4}\,d^{-3/16}. \qquad \square$$

B.4 Category Disjointness

Definition 3.1 requires the categories to be disjoint. The random construction of Section 3.1 delivers this except with exponentially small probability; the intuition is that an independent random partition almost never leaves a two-signed vector block-constant. We first record a uniform Chernoff estimate for the pinned parts, which strengthens the per-category Chebyshev bound of Lemma B.6 to an exponentially small failure probability (as needed both below and in the proof of Corollary 3.4, where a bound holding simultaneously for all N categories is required and a per-category $1-\delta$ guarantee would be vacuous after the union bound).

Lemma B.7 (Uniform pin-count bound)**.** *$|\pi_{l,0}| \sim \mathrm{Binomial}(d, \varepsilon/\sqrt{d})$ for each l, so by the multiplicative Chernoff bound, $\mathbb{P}[|\pi_{l,0}| > 2\varepsilon\sqrt{d}] \le e^{-\varepsilon\sqrt{d}/3}$, and*

$$\mathbb{P}\big[\exists\, l \in [N] :\ |\pi_{l,0}| > 2\varepsilon\sqrt{d}\big] \le N e^{-\varepsilon\sqrt{d}/3}.$$

Lemma B.8 (Category disjointness)**.** *Under the hypotheses of Theorem 3.2 (in particular $d = \Omega(n^8)$, $n = \Omega(1/\varepsilon^2)$, and $N = n^2/\delta$ with $\delta^{-1} \in \mathrm{poly}(n)$), with probability $1 - e^{-\Omega(\varepsilon\sqrt{d})}$ over the partitions $\Pi_1, \dots, \Pi_N$, the categories $C_1, \dots, C_N$ are pairwise disjoint.*

Proof. Condition on two events. First, every block of every partition has size in $[d^{3/4}/2,\, 2d^{3/4}]$: each $|\pi_{l,k}|$ is $\mathrm{Binomial}(d - |\pi_{l,0}|,\, d^{-1/4})$ with mean $(1 \pm o(1))d^{3/4}$, so by Chernoff and a union bound over the $Nd^{1/4}$ blocks this fails with probability at most $2Nd^{1/4}e^{-\Omega(d^{3/4})}$. Second, $|\pi_{l,0}| \le 2\varepsilon\sqrt{d}$ for every l, which fails with probability at most $Ne^{-\varepsilon\sqrt{d}/3}$ (Lemma B.7).

Fix $l \ne l'$ and condition on all pinned parts $\{\pi_{m,0}\}_{m\in[N]}$ and all pin values (the pin values are a function of the pinned parts and μ, and, jointly with Π_l's bin assignment, they determine C_l), as well as on Π_l's bin assignment; then fix $z \in C_l$. The only randomness used below is the bin assignment of $\Pi_{l'}$'s unpinned coordinates, which remains i.i.d. uniform and independent of everything conditioned on, and the counting bounds that follow hold uniformly over every conditioned realization on the two events above. Since the block signs of z do not all coincide (the two such vectors are excluded from C_l), it has at least one block of each sign under Π_l, hence at least $m \ge d^{3/4}/2$ coordinates carrying its minority sign, and at least $(d - |\pi_{l,0}|)/2 \ge d/4$ coordinates carrying its majority sign. For $z \in C_{l'}$ it is necessary that every coordinate of z is either pinned in $\pi_{l',0}$ (with matching pin value – a requirement we do not even need, as pins only make overlap harder) or lies in a block of $\Pi_{l'}$ that is monochromatic under z. Since $|\pi_{l',0}| \le 2\varepsilon\sqrt{d} \le m/2$ for d large, at least $m/2 \ge d^{3/4}/4$ minority coordinates of z are unpinned in $\Pi_{l'}$, and each must then lie in a $\Pi_{l'}$-block containing no majority coordinate of z. But at least $d/4 - 2\varepsilon\sqrt{d} \ge d/8$ majority coordinates are unpinned in $\Pi_{l'}$ and are assigned to the $d^{1/4}$ bins independently and uniformly, so the probability that some fixed bin receives none of them is at most $(1 - d^{-1/4})^{d/8} \le e^{-d^{3/4}/8}$. By a union bound over bins, with probability at least $1 - d^{1/4}e^{-d^{3/4}/8}$ every bin of $\Pi_{l'}$ contains a majority coordinate of z; on that event any unpinned minority coordinate of z sits in a non-monochromatic block, so $z \notin C_{l'}$.

A union bound over the at most $2^{d^{1/4}}$ elements of C_l and the N^2 ordered pairs (l, l') bounds the total overlap probability by

$$N^2\, 2^{d^{1/4}}\, d^{1/4} e^{-d^{3/4}/8} \;+\; 2Nd^{1/4}e^{-\Omega(d^{3/4})} \;+\; Ne^{-\varepsilon\sqrt{d}/3} \;=\; e^{-\Omega(\varepsilon\sqrt{d})},$$

where the last equality holds because $\varepsilon\sqrt{d}$ dominates both $d^{1/4}$ and $\log N$ at the dimension of Theorem 3.2 ($\varepsilon\sqrt{d} \geq \varepsilon d^{1/4} \cdot d^{1/4}$ with $\varepsilon d^{1/4} = \Omega(n^{3/2})$ when $d = \Omega(n^8)$ and $n = \Omega(1/\varepsilon^2)$). □

Remark B.9 (Deterministic trimming). *Disjointness can also be enforced deterministically: redefine* $C_l' := C_l \setminus \bigcup_{l'<l} C_{l'}$. *The trimmed categories are disjoint by construction, and Algorithm 2 is unaffected, since its candidate for category* l *is built from* Π_l *alone. By Lemma B.8, nothing is trimmed except with probability* $e^{-\Omega(\varepsilon\sqrt{d})}$, *so all tail bounds carry over up to that additive term.*

C DEFERRED PROOFS

C.1 Soundness Transfer to Δ_μ

Proof of Lemma 3.3. Let $\bar{C}_l$ denote the category with the two excluded vectors (those whose block signs all coincide) added back; each $z \in \bar{C}_l$ is parameterized by independent Rademachers $s_1, \ldots, s_{d^{1/4}}$ (the block signs), and the inner product becomes $\sum_k s_k c_{l,k}$ where $c_{l,k} = \frac{1}{\sqrt{d}} \sum_{i \in \pi_{l,k}} y_i$. By Cauchy–Schwarz: $\sum_k c_{l,k}^2 \leq \frac{M_l}{d}\|y\|^2$. Applying Hoeffding's inequality to the sum of independent bounded terms $s_k c_{l,k}$ with $\sum_k c_{l,k}^2 \leq M_l\|y\|^2/d$ bounds the tail under Uniform($\bar{C}_l$) by $\exp(-t^2 d/(2M_l\|y\|^2))$. Conditioning on avoiding the excluded pair inflates any event's probability by at most $(1 - 2^{1-d^{1/4}})^{-1} \leq 2$ and leaves $\mathbb{E}[z \mid l]$ unchanged (the pair is sign-flip symmetric), giving the stated bound. □

Lemma C.1 (Center-shift bound). *Let* $\mu \in \mathbb{R}^d$ *be an arbitrary fixed centering vector. Fix* $y \in \mathbb{R}^d$ *independent of a fresh draw* $Z_0 \sim \Delta_\mu$. *If* $|\langle y, \mathbb{E}[z \mid l] - \mu\rangle| \leq \Delta_{\text{shift}}$ *for all* $l \in [N]$, *then for* $t \geq \Delta_{\text{shift}}$:

$$\mathbb{P}[\langle y, Z_0 - \mu\rangle \geq t] \leq 2\exp\left(-\frac{(t - \Delta_{\text{shift}})^2 d}{2M\|y\|^2}\right),$$

where $M = \max_l \max_k |\pi_{l,k}|$.

Proof. Condition on l and decompose: $\langle y, Z_0 - \mu\rangle = \langle y, Z_0 - \mathbb{E}[z \mid l]\rangle + \langle y, \mathbb{E}[z \mid l] - \mu\rangle$. The second term is bounded by Δ_{shift}, so if the LHS $\geq t$, then $\langle y, Z_0 - \mathbb{E}[z \mid l]\rangle \geq t - \Delta_{\text{shift}}$. Apply the category tail bound (Lemma 3.3) conditionally and average over l. □

Proof of Corollary 3.4. Set $y = A\hat{\theta}$ where $A = \text{diag}(g(\varepsilon, \mu))$. Since $\|A\|_2 \leq 288\varepsilon^2$ and $\|\hat{\theta}\| \leq 1$, we have $\|y\| \leq 288\varepsilon^2$.

We bound the shift $\Delta_{\text{shift}} = \max_l |\langle A\hat{\theta}, \mathbb{E}[z \mid l] - \mu\rangle|$ by splitting the coordinates by pinned status. For $i \notin \pi_{l,0}$ we have $\mathbb{E}[z_i \mid l] = 0$, so these coordinates contribute at most $|\sum_{i \notin \pi_{l,0}} g_i \hat{\theta}_i \mu_i| \leq \|g \odot \mu\|\|\hat{\theta}\| \leq 3456\varepsilon^3$ – the same Cauchy–Schwarz estimate as the μ-term in Step 2 of Appendix C.3 – which is at most $\varepsilon/(4n)$ since $n \leq 1/(13824\,\varepsilon^2)$. For $i \in \pi_{l,0}$ we have $|\mathbb{E}[z_i \mid l] - \mu_i| = |B_{i,l}/\sqrt{d} - \mu_i| \leq 2/\sqrt{d}$, so by Cauchy–Schwarz (as in Lemma B.6) these coordinates contribute at most $288\varepsilon^2 \cdot (2/\sqrt{d}) \cdot \sqrt{|\pi_{l,0}|}$. Since the bound must hold simultaneously for all N categories, we use the uniform pin-count bound (Lemma B.7) rather than the per-category Chebyshev estimate: except with probability $Ne^{-\varepsilon\sqrt{d}/3}$ – which is $O(\delta^2/n^2)$ at the theorem's dimension – every category has $|\pi_{l,0}| \leq 2\varepsilon\sqrt{d}$, so the pinned contribution is at most $576\sqrt{2}\,\varepsilon^{5/2} d^{-1/4}$, itself at most $\varepsilon/(8n)$ at the theorem's dimension. Hence $\Delta_{\text{shift}} \leq \varepsilon/(4n) + \varepsilon/(8n)$, and at the threshold $t = \varepsilon/(2n)$ the budget $t - \Delta_{\text{shift}} \geq \varepsilon/(8n)$ survives.

By Lemma C.1 with $t = \varepsilon/(2n)$, and a union bound over the n fresh samples:

$$\mathbb{P}[\exists i : \langle A\hat{\theta}, Z_i - \mu\rangle \geq \varepsilon/(2n)] \leq 2n\exp\left(-\frac{(\varepsilon/8n)^2 d}{2M \cdot 288^2\varepsilon^4}\right) = 2n\exp\left(-\frac{d}{2^7 \cdot 288^2\, M\varepsilon^2 n^2}\right).$$

With $d = \Omega(M \cdot n^2 \log(n/\delta))$, this is $\leq O(\delta^2/n^2)$. □

C.2 Mean-Alignment via Bit-Fitting

Proposition C.2 (Mean-alignment error). *The per-coordinate mean error satisfies* $|\mathbb{E}_{z \sim \Delta_\mu}[z_i] - \mu_i| \leq 1/(N\sqrt{d})$.

Proof. $\mathbb{E}[z_i] = \frac{1}{N}\sum_{l\in K_i} B_{i,l}/\sqrt{d}$ (non-pinned categories contribute 0 by symmetry). Bit-fitting minimizes the distance to μ_i, and adjusting one bit changes the sum by $2/(N\sqrt{d})$, so the error is at most $1/(N\sqrt{d})$. □

C.3 Complete Sign Consistency Derivation

Full proof of Lemma B.4. Let z be a training datapoint from category C_l with block signs $(s_1, \ldots, s_{d^{1/4}})$, satisfying $\langle v(z), \hat{\theta}\rangle \geq \varepsilon/n$.

Step 1: Block decomposition. Expanding the inner product:

$$\begin{aligned}\langle v(z), \hat{\theta}\rangle &= \underbrace{\sum_{i\in\pi_{l,0}} g_i(z_i - \mu_i)\hat{\theta}_i}_{S_0} + \sum_{k=1}^{d^{1/4}} \sum_{i\in\pi_{l,k}} g_i\left(\frac{s_k}{\sqrt{d}} - \mu_i\right)\hat{\theta}_i \\ &= S_0 + \sum_{k=1}^{d^{1/4}} \left(\frac{s_k}{\sqrt{d}} S_{l,k} - \sum_{i\in\pi_{l,k}} g_i\mu_i\hat{\theta}_i\right),\end{aligned}$$

where $g_i = g(\varepsilon,\mu)_i$ and $S_{l,k} = \sum_{i\in\pi_{l,k}} g_i\hat{\theta}_i$.

Step 2: Bounding error terms. The μ-term is the residual bias that appears when replacing z_i by the block sign $s_k/\sqrt{d}$ in non-pinned blocks: since $z_i - \mu_i = (s_k/\sqrt{d} - \mu_i)$, each block contributes an extra $-\sum_{i\in\pi_{l,k}} g_i\mu_i\hat{\theta}_i$. Summing over all blocks gives the total μ-term $\sum_i g_i\mu_i\hat{\theta}_i$, which we bound by Cauchy–Schwarz: $|\sum_i g_i\mu_i\hat{\theta}_i| \leq \|g\odot\mu\|\|\hat{\theta}\| \leq \|g\odot\mu\| \leq \sqrt{\sum_i (288\varepsilon^2)^2\cdot(12\varepsilon)^2/d} = 288\cdot 12\varepsilon^3 = 3456\varepsilon^3$. Since $n \leq 1/(13824\,\varepsilon^2)$, this is at most $\varepsilon/(4n)$. (This is the one error term that does not vanish as d grows, and it is what fixes the upper end of the regime $n = \Theta(1/\varepsilon^2)$.)

The pinned term: by Lemma B.6, with probability $1-\delta$ over the partition, $|S_0| \leq 288\varepsilon^2\cdot(2/\sqrt{d})\cdot\sqrt{|\pi_{l,0}|} \leq 576\sqrt{2}\,\varepsilon^{5/2}\,\delta^{-1/4}\,d^{-3/16}$. As $\delta^{-1} \in \mathrm{poly}(n)$, this decays with d and is below $\varepsilon/(4n)$ at the theorem's dimension (it is dominated by the $\max\{\varepsilon^4 n^4/\delta^2,\ n^8\log^4(n/\delta)\}$ terms of d_0).

So:

$$\frac{1}{\sqrt{d}}\sum_k s_k S_{l,k} \geq \frac{\varepsilon}{n} - \frac{\varepsilon}{4n} - \frac{\varepsilon}{4n} = \frac{\varepsilon}{2n}.$$

Step 3: Per-block concentration. The average signed block score is $\geq \varepsilon\sqrt{d}/(2n\cdot d^{1/4}) = \varepsilon d^{1/4}/(2n)$.

For the random partition, $S_{l,k} = \sum_{i\notin\pi_{l,0}} g_i\hat{\theta}_i\mathbb{1}[i\in\pi_{l,k}]$ where each indicator is independent with $\mathbb{P}[i\in\pi_{l,k}] = d^{-1/4}$. By Lemma B.5, $\mathrm{Var}(S_{l,k}) \leq 288^2\varepsilon^4 d^{-1/4}$, so $\sigma_k \leq 288\varepsilon^2 d^{-1/8}$.

Chebyshev with $\kappa = \sqrt{d^{1/4}/\delta}$:

$$\mathbb{P}[|S_{l,k} - \mathbb{E}[S_{l,k}]| \geq \kappa\sigma_k] \leq \delta/d^{1/4}.$$

The deviation $\kappa\sigma_k = 288\varepsilon^2 d^{1/8}\cdot d^{-1/8}/\sqrt{\delta} = 288\varepsilon^2/\sqrt{\delta}$ is dimension-free, and is smaller than the signal $\varepsilon d^{1/4}/(2n)$ exactly when

$$d^{1/4} \geq \frac{288\varepsilon n}{\sqrt{\delta}}\cdot 2 = \frac{576\varepsilon n}{\sqrt{\delta}},$$

i.e., $d \geq \Omega(\varepsilon^4 n^4/\delta^2)$ – which is precisely the hypothesis of Lemma B.4. Under this condition, each $s_k S_{l,k} > 0$.

Step 4: Union bound. By the union bound over all $d^{1/4}$ blocks, sign consistency holds simultaneously with probability $\geq 1-\delta$. □

C.4 Reweighting Norm Bound

Proof of Lemma B.3. $g(\varepsilon,\mu)_i^2 = ((144\varepsilon^2 - d\mu_i^2)/(1-d\mu_i^2))^2 \leq (144\varepsilon^2/(1-144\varepsilon^2))^2 \leq (288\varepsilon^2)^2$, using $d\mu_i^2 \leq 144\varepsilon^2$ and $\varepsilon \leq 1/(12\sqrt{2})$. Sum over J for the norm bound. □

C.5 Complete Proof of Theorem 3.2

Proof of Theorem 3.2. We assemble the individual lemma results. Fix $\varepsilon \le \varepsilon_0 = 1/(12\sqrt{2})$, $\delta \le \varepsilon$ with $\delta^{-1} \in \mathrm{poly}(n)$, $n \ge \Omega(1/\varepsilon^2)$, and $d \ge \Omega(\max\{n^8 \log^4(n/\delta),\ \varepsilon^4 n^4/\delta^2,\ n^2 \log(n+\delta^{-1})\})$ (by Remark D.1, the M-dependent soundness constraint resolves to the $n^8 \log^4$ term).

Step 1: Distribution construction. The block-uniform distribution Δ_μ is constructed as in Section 3.1: partition sampling (Algorithm 3), bit-fitting (Algorithm 4), and category definition yield $N = n^2/\delta$ categories, each of size $2^{d^{1/4}} - 2$ (on the event that every block is non-empty, which fails with probability at most $d^{1/4} e^{-\Omega(d^{3/4})}$ and is folded into the failure events below), pairwise disjoint except with probability $e^{-\Omega(\varepsilon\sqrt{d})}$ (Lemma B.8). By Proposition C.2, the per-coordinate mean error satisfies $|\mathbb{E}_{z\sim\Delta_\mu}[z_i] - \mu_i| \le 1/(N\sqrt{d})$.

Step 2: Correlation bound. By Lemma B.2, under the uniform prior $\mu \sim U = \mathrm{Uniform}([-12\varepsilon/\sqrt{d}, 12\varepsilon/\sqrt{d}]^d)$, the expected total reweighted correlation satisfies

$$\mathbb{E}_{\mu,S_n}\left[\sum_{j=1}^{n} \langle v(z_j), \hat{\theta}\rangle\right] \ge 6\varepsilon - 4\delta \ge 2\varepsilon,$$

where the last inequality uses $\delta \le \varepsilon$. The correlation bound is derived from the distribution-free excess risk guarantee of the ε-learner. Because excess risk is bounded for *all* distributions – including Δ_μ, whose mean satisfies $\|\mathbb{E}_{\Delta_\mu}[z] - \mu\|_\infty \le 1/(N\sqrt{d})$ – the accuracy-correlation relationship (Lemma D.3) applies to Δ_μ as well, with an additional error of at most $\sqrt{d} \cdot 1/(N\sqrt{d}) = 1/N = O(\delta/n^2)$, which is negligible.

By Markov's inequality on the complement, with probability $\ge 1-\delta$ over (μ, S_n), at least $\Omega(n)$ training datapoints satisfy $\langle v(z_j), \hat{\theta}\rangle \ge \varepsilon/n$.

Step 3: Soundness (no false positives). For a fresh sample $Z_0 \sim \Delta_\mu$ not in the training set, Corollary 3.4 (proved via the category tail bound, Lemma 3.3, and the center-shift bound, Lemma C.1) gives

$$\mathbb{P}[\langle g(\varepsilon,\mu) \odot (Z_0 - \mu), \hat{\theta}\rangle \ge \varepsilon/n] \le \exp\left(-\Omega\left(\frac{d}{M\varepsilon^2 n^2}\right)\right)$$

(the corollary in fact proves this at the smaller threshold $\varepsilon/(2n)$, which implies the bound at the test's threshold ε/n). With $d = \Omega(M \cdot n^2 \log(n/\delta))$, this probability is at most $O(\delta^2/n^2)$. A union bound over $N = n^2/\delta$ categories gives a total false-positive probability of at most $N \cdot O(\delta^2/n^2) = O(\delta)$.

Step 4: Exact reconstruction via sign consistency. For each training datapoint z from category C_l satisfying $\langle v(z), \hat{\theta}\rangle \ge \varepsilon/n$, Lemma B.4 guarantees that $\mathrm{sign}(S_{l,k}) = s_k$ for all blocks $k \in [d^{1/4}]$ simultaneously, with probability $\ge 1 - O(\delta)$ over the random partition. This requires $d \ge \Omega(\varepsilon^4 n^4/\delta^2)$. Since pinned coordinates are recovered by construction ($z'_i = B_{i,l}/\sqrt{d} = z_i$ for $i \in \pi_{l,0}$), the algorithm reconstructs z exactly.

Step 5: Union bound and conclusion. Define the following events:

- E_0: the categories are pairwise disjoint (Step 1; fails with probability $e^{-\Omega(\varepsilon\sqrt{d})}$ by Lemma B.8, absorbed into $O(\delta)$ at the theorem's dimension),
- E_1: at least $\Omega(n)$ training datapoints have $\langle v(z_j), \hat{\theta}\rangle \ge \varepsilon/n$ (Step 2),
- E_2: no false positives among the N categories (Step 3),
- E_3: sign consistency holds for all training datapoints with large correlation (Step 4).

Each event fails with probability at most $O(\delta)$. By a union bound, all four hold simultaneously with probability $\ge 1 - O(\delta)$. Conditioning on $E_0 \cap E_1 \cap E_2 \cap E_3$, Algorithm 2 reconstructs $\Omega(n)$ training datapoints exactly and produces no false positives. The algorithm iterates over $N = n^2/\delta$ categories, performing $O(d)$ work per category, for a total runtime of $O(Nd) = O(n^2 d/\delta) = \mathrm{poly}(n, d, \delta^{-1})$. □

C.6 Correlation Bound

Proof of Lemma B.2. By the accuracy-correlation relationship (Lemma D.3): $\langle \mathbb{E}[\hat{\theta}], \mu \rangle \geq \|\mu\| - \varepsilon - 2\delta$. Under the prior U, $\mathbb{E}_\mu[\|\mu\|] \geq 4\varepsilon$ (Lemma D.2), so $\mathbb{E}_\mu[\langle \mathbb{E}[\hat{\theta}], \mu \rangle] \geq 3\varepsilon - 2\delta$.

The reweighting vector $v(z)_i = g(\varepsilon, \mu)_i (z_i - \mu_i)$ amplifies the correlation because $\mathbb{E}[v(z)_i \hat{\theta}_i] = g(\varepsilon, \mu)_i \cdot \mathrm{Cov}(z_i, \hat{\theta}_i)$. Since $d\mu_i^2 \leq 144\varepsilon^2$ by the parameter constraint, we have

$$g(\varepsilon, \mu)_i = \frac{144\varepsilon^2 - d\mu_i^2}{1 - d\mu_i^2} \geq 0,$$

with $g(\varepsilon, \mu)_i = 144\varepsilon^2$ at $\mu_i = 0$ and $g(\varepsilon, \mu)_i = 0$ at $d\mu_i^2 = 144\varepsilon^2$. Although $g(\varepsilon, \mu)_i$ may be smaller than 1 for small ε, the reweighting redistributes weight towards coordinates with small $|\mu_i|$ (where $g_i \approx 144\varepsilon^2$) and away from coordinates with large $|\mu_i|$ (where $g_i \approx 0$). Following the argument in Attias et al. (2024, Lemma 6.3), summing the reweighted correlation over all coordinates and all n training datapoints yields $\mathbb{E}[\sum_{j=1}^n \langle v(z_j), \hat{\theta} \rangle] \geq 6\varepsilon - 4\delta$. □

D ADDITIONAL TECHNICAL DETAILS

D.1 Notation Summary

Table 1: Summary of notation

Symbol	Meaning
d	Dimension of parameter space
n	Sample size
N	Number of candidates (Definition 2.2) and of categories; $N = n^2/\delta$
$\mathcal{D}_\theta$, Δ_θ	Ambient family; hard distribution Uniform(Σ), $\Sigma \sim \mathcal{D}_\theta^{\otimes N}$ (Definition 2.2)
ε, δ	Excess risk and confidence parameters
$\mathcal{Z}$	Sample domain ($\{\pm 1/\sqrt{d}\}^d$); uppercase Z_i denotes a random datapoint
$\mathcal{D}_\mu$	Product distribution over $\mathcal{Z}$ with $\mathbb{E}[z_i] = \mu_i$
Δ_μ	Block-uniform hard distribution (Definition 3.1), not a product
$\hat{\theta}$	Learned parameter ($\mathcal{A}_n(S_n)$)
$\pi_{l,0}$	Mean-alignment part of partition Π_l
$\pi_{l,k}$	Block k of partition Π_l
$B_{i,l}$	Pin value for coordinate i in category l
$g(\varepsilon, \mu)$	Reweighting vector
$v(z)$	Weighted correlation vector
$\odot$	Hadamard (entrywise) product
$S_{l,k}$	Block score: $\sum_{i \in \pi_{l,k}} g_i \hat{\theta}_i$

D.2 Parameter Relationships

The main theorems require: (1) $n = \Theta(1/\varepsilon^2)$, i.e. $\Omega(1/\varepsilon^2) \leq n \leq 1/(13824\,\varepsilon^2)$, the regime of Theorem 3.2: the lower end is the sample-complexity threshold inherited from the membership-inference lower bound of Attias et al. (2024), and the upper end is fixed by the μ-term budget in the sign-consistency and soundness steps (Appendix C.3; Corollary 3.4); (2) $d \geq \max\{\Omega(n^8 \log^4(n/\delta)),\ \Omega(\varepsilon^4 n^4/\delta^2),\ \Omega(n^2 \log(n + \delta^{-1}))\}$ (the first term resolves the self-referential block-size constraint of Remark D.1 below: each non-pinned coordinate is assigned uniformly to one of $d^{1/4}$ blocks, so the expected block size is $(d - |\pi_{l,0}|)/d^{1/4} \approx d^{3/4}$ and standard concentration gives $M = O(d^{3/4})$ with high probability; substituting into $d = \Omega(M \cdot n^2 \log(n/\delta))$ yields $d^{1/4} = \Omega(n^2 \log(n/\delta))$, i.e. $d = \Omega(n^8 \log^4(n/\delta))$); (3) $\delta \leq \varepsilon$, $\delta^{-1} \in \mathrm{poly}(n)$; (4) $N = n^2/\delta$; (5) $\mu \in [-12\varepsilon/\sqrt{d}, 12\varepsilon/\sqrt{d}]^d$.

Remark D.1 (Resolving the dimension requirement)**.** *The dimension requirement in Theorem 3.2 is the maximum of three independent constraints: soundness ($d = \Omega(n^8 \log^4(n/\delta))$: the requirement $d = \Omega(M \cdot n^2 \log(n/\delta))$ of Corollary 3.4 is self-referential, since the maximum block size $M = \max_l \max_k |\pi_{l,k}|$ depends on d, and it resolves via $M = O(d^{3/4})$ with high probability; see item (2) above), sign consistency ($d = \Omega(\varepsilon^4 n^4/\delta^2)$, Lemma B.4), and*

mean alignment ($d = \Omega(n^2 \log(n + \delta^{-1}))$). Under the theorem's hypotheses with $\delta^{-1} = O(n^3)$ – in particular for any δ^{-1} of low polynomial degree – soundness dominates; for polynomially smaller δ the sign-consistency term $\varepsilon^4 n^4/\delta^2$ takes over. All three are absorbed by the single $\Omega(\cdot)$ in the theorem statement.

D.3 Supplementary Lemmas

Lemma D.2 (Expected norm of uniform prior)**.** *For $U = Uniform([-12\varepsilon/\sqrt{d}, 12\varepsilon/\sqrt{d}]^d)$, $\mathbb{E}_{\mu\sim U}[\|\mu\|] \geq 4\varepsilon$.*

Proof. $\mathbb{E}[\|\mu\|^2] = d \cdot 144\varepsilon^2/(3d) = 48\varepsilon^2$, and $\|\mu\| \leq \sqrt{d} \cdot 12\varepsilon/\sqrt{d} = 12\varepsilon$ almost surely, so $\|\mu\|^2 \leq 12\varepsilon \|\mu\|$ and $\mathbb{E}[\|\mu\|] \geq \mathbb{E}[\|\mu\|^2]/(12\varepsilon) = 4\varepsilon$. □

Lemma D.3 (Accuracy-correlation relationship)**.** *For an ε-learner and $f(\theta, z) = -\langle\theta, z\rangle$: $\langle\mathbb{E}[\hat{\theta}], \mu\rangle \geq \|\mu\| - \varepsilon - 2\delta$.*

Proof. The optimal $\theta^* = \mu/\|\mu\|$ achieves risk $-\|\mu\|$. Excess risk $\leq \varepsilon$ gives $\langle\hat{\theta}, \mu\rangle \geq \|\mu\| - \varepsilon$ with probability at least $1 - \delta$ over $S_n \sim \mathcal{D}_\mu^{\otimes n}$. Decomposing the expectation over the high-probability event and its complement:

$$\begin{aligned}
\langle\mathbb{E}[\hat{\theta}], \mu\rangle &= \mathbb{E}[\langle\hat{\theta}, \mu\rangle] \\
&= \mathbb{P}[E] \cdot \mathbb{E}[\langle\hat{\theta}, \mu\rangle \mid E] + \mathbb{P}[E^c] \cdot \mathbb{E}[\langle\hat{\theta}, \mu\rangle \mid E^c] \\
&\geq (1 - \delta)(\|\mu\| - \varepsilon) + \delta \cdot (-\|\mu\|) \\
&= \|\mu\| - \varepsilon - 2\delta\|\mu\| \\
&\geq \|\mu\| - \varepsilon - 2\delta,
\end{aligned}$$

where E denotes the event that excess risk $\leq \varepsilon$ holds, the third line uses $\langle\hat{\theta}, \mu\rangle \geq -\|\hat{\theta}\|\|\mu\| \geq -\|\mu\|$ (since $\|\hat{\theta}\| \leq 1$), and the last step uses $\|\mu\| \leq 1$. □

E GENERALIZATIONS, BARRIERS, AND OPEN DIRECTIONS

This appendix collects the discussion of the adversarial model and of the reach of the framework: which parts of the model are necessary, which relax, and where the bottlenecks to generalization lie. Appendix E.1 shows that the adversary's knowledge of the distribution parameter relaxes to sampling access at polynomial cost. Appendix E.2 shows that the remaining feature of the model – a hard distribution chosen per learner – is necessary at the standard sample size. Appendix E.3 isolates the properties of the category construction that the exponential-domain argument actually uses. Appendices E.4 and E.5 discuss continuous distributions and the exponential-family extension we regard as the main open problem.

E.1 Sampling Access Suffices

The adversary of Theorem 3.2 receives the realized mean μ. This is a convenience, not a load-bearing assumption: it relaxes to sampling access at the realized parameter – the resource practical membership inference attacks assume (Shokri et al., 2017; Carlini et al., 2022) – at polynomial cost. The game is otherwise unchanged: the family, the prior, and the partition-sampling and bit-fitting rules stay fixed and public, and the partitions Π_l are the adversary's own coins; only μ is withheld.

Proposition E.1 (Sampling access suffices)**.** *Consider the setting of Theorem 3.2, with the game modified so that the realized μ is withheld from the adversary, which instead receives m_1 fresh i.i.d. samples from the product distribution $\mathcal{D}_\mu$ and m_2 fresh i.i.d. samples from Δ_μ, drawn independently of the training set. There is an absolute constant C such that whenever*

$$m_1 \;\geq\; C\, n^2 d^{1/4} \log(d/\delta) \qquad \text{and} \qquad m_2 \;\geq\; C\, N \log(N/\delta),$$

the conclusion of Theorem 3.2 continues to hold with probability $1 - O(\delta)$ for the adversary that runs Algorithm 2 with the projected empirical estimate $\hat{\mu}$ in place of μ, the pin values read off the Δ_μ-samples, and acceptance threshold $3\varepsilon/(4n)$ in place of ε/n. The total budget is thus $m = \Omega(n^2 d^{1/4} + N \log N)$ up to the logarithmic factors – polynomial in n, d, and δ^{-1} – and we have made no attempt to optimize it.

Proof. Throughout, condition on the events of the original analysis (correlation, soundness, sign consistency, disjointness), which are unaffected by the extra samples. Write $\Delta\mu = \hat{\mu} - \mu$, $g = g(\varepsilon, \mu)$, $\hat{g} = g(\varepsilon, \hat{\mu})$, and $\Delta g = \hat{g} - g$.

Step 1: Estimation and projection. Let $\hat{\mu}$ be the coordinate-wise empirical mean of the m_1 samples from $\mathcal{D}_\mu$, projected onto the box $[-12\varepsilon/\sqrt{d}, 12\varepsilon/\sqrt{d}]^d$. Since μ lies in the box, the projection can only decrease each $|\Delta\mu_i|$, and it keeps the bounds of Lemma B.3 valid for $\hat{g}$. Each coordinate satisfies $z_i \in \{\pm 1/\sqrt{d}\}$, so by Hoeffding's inequality and a union bound over the d coordinates, with probability $1 - \delta$,

$$|\Delta\mu_i| \;\le\; \tau := \sqrt{\frac{2\log(4d/\delta)}{m_1 d}} \quad \text{for every } i \in [d],$$

hence $\|\Delta\mu_J\|_2 \le \sqrt{|J|}\,\tau$ for every $J \subseteq [d]$.

Step 2: Perturbation of the statistic. Differentiating $g_i = (144\varepsilon^2 - d\mu_i^2)/(1 - d\mu_i^2)$ under the constraint $d\mu_i^2 \le 144\varepsilon^2$ gives $|\partial g_i/\partial \mu_i| \le 2d\,|\mu_i|\,(1 - 144\varepsilon^2)^{-2} \le 96\varepsilon\sqrt{d}$ for $\varepsilon \le \varepsilon_0$, so $|\Delta g_i| \le 96\varepsilon\sqrt{d}\,|\Delta\mu_i|$. For any $x \in \mathcal{Z}$ (so $|x_i - \hat{\mu}_i| \le 2/\sqrt{d}$ after projection), the adversary's statistic decomposes as

$$\hat{g} \odot (x - \hat{\mu}) - g \odot (x - \mu) \;=\; \Delta g \odot (x - \hat{\mu}) \;-\; g \odot \Delta\mu,$$

whose two parts contribute, after pairing with $\|\hat{\theta}\| \le 1$,

$$|\langle \Delta g \odot (x - \hat{\mu}), \hat{\theta}\rangle| \le \tfrac{2}{\sqrt{d}}\|\Delta g\|_2 \le 192\varepsilon\|\Delta\mu\|_2, \qquad |\langle g \odot \Delta\mu, \hat{\theta}\rangle| \le \|g\|_\infty \|\Delta\mu\|_2 \le 288\varepsilon^2\|\Delta\mu\|_2.$$

These are two additive parts of one and the same difference, so they share a single $\varepsilon/(8n)$ estimation allotment, half each: it suffices that $\|\Delta\mu\|_2 \le 1/(3072\,n)$ and $\|\Delta\mu\|_2 \le 1/(4608\,\varepsilon n)$, i.e. (via $\|\Delta\mu\|_2 \le \sqrt{d}\,\tau$) that $m_1 = \Omega(n^2\log(d/\delta))$ and $m_1 = \Omega(\varepsilon^2 n^2 \log(d/\delta))$ respectively – both dominated by the block-score constraint of Step 3. Hence the statistic of every candidate moves by at most $\varepsilon/(8n)$.

Step 3: Perturbation of the block scores (the binding constraint). The decoder reads $\operatorname{sign}(\hat{S}_{l,k})$ with $\hat{S}_{l,k} = \sum_{i \in \pi_{l,k}} \hat{g}_i \hat{\theta}_i$. By Cauchy–Schwarz and Step 1,

$$\begin{aligned} |\hat{S}_{l,k} - S_{l,k}| \;&\le\; \|\Delta g_{\pi_{l,k}}\|_2\,\|\hat{\theta}\| \;\le\; 96\varepsilon\sqrt{d}\cdot\sqrt{|\pi_{l,k}|}\,\tau \\ &\le\; 96\varepsilon\sqrt{d}\cdot\sqrt{2d^{3/4}}\,\tau \;=\; O\!\left(\varepsilon\, d^{3/8}\sqrt{\tfrac{\log(d/\delta)}{m_1}}\right), \end{aligned}$$

using the block-size bound $|\pi_{l,k}| \le 2d^{3/4}$ from the proof of Lemma B.8. The signal is $\varepsilon d^{1/4}/(2n)$ (Step 3 of Appendix C.3); requiring the perturbation to be at most $\varepsilon d^{1/4}/(8n)$ forces $m_1 = \Omega(n^2 d^{1/4}\log(d/\delta))$ – the binding constraint, and the one that grows with d. Sign consistency then survives: the Chebyshev deviation $288\varepsilon^2/\sqrt{\delta}$ plus the perturbation $\varepsilon d^{1/4}/(8n)$ stay below the signal exactly when $d^{1/4} \ge 768\varepsilon n/\sqrt{\delta}$, the same asymptotics as Lemma B.4.

Step 4: Recall and soundness at the shifted threshold. By Step 2, every candidate's statistic moves by at most $\varepsilon/(8n)$. A training point flagged by the exact analysis has true statistic $\ge \varepsilon/n$, hence perturbed statistic $\ge \varepsilon/n - \varepsilon/(8n) \ge 3\varepsilon/(4n)$: it passes the modified threshold. Conversely, a fresh point passing the modified threshold has true statistic $\ge 3\varepsilon/(4n) - \varepsilon/(8n) = 5\varepsilon/(8n) \ge \varepsilon/(2n)$, so the tail of Corollary 3.4 – proven at threshold $\varepsilon/(2n)$ – applies verbatim and the false-positive analysis of Appendix C.5 goes through unchanged. This is why the relaxation runs in the same regime $n \le 1/(13824\,\varepsilon^2)$ as the theorem: the detection margin $\varepsilon/(2n)$ and the soundness slack are already budgeted there, and estimation consumes only the residual $\varepsilon/(8n)$.

Step 5: Pins by coverage. The pin values are the one discrete function of μ, and the adversary does not recompute them from $\hat{\mu}$: it reads them off the fresh Δ_μ-samples. Each such sample belongs to its category l and is block-constant on the non-pinned part of Π_l; by the argument of Lemma B.8 (which never uses the pin values), it is block-constant on no other partition, except with probability $e^{-\Omega(\varepsilon\sqrt{d})}$. The adversary therefore classifies each sample by checking block-constancy against its own partitions, and within a category the pinned coordinates are constant across samples, so a single sample of category l reveals all of $(B_{i,l})_{i \in \pi_{l,0}}$. Since Δ_μ draws $l \sim \text{Uniform}([N])$, covering the categories is a coupon-collector problem: $m_2 \ge CN\log(N/\delta)$ covers every

category with probability $1-\delta$. ($m_2 = \Omega(N)$ already covers a constant fraction of the categories, which preserves the $\Omega(n)$ yield since the training points' categories are uniform as well; we state the full-coverage version for simplicity.)

A union bound over the failure events of Steps 1–5 and of the original analysis completes the proof. □

E.2 Necessity of a Per-Learner Hard Distribution

The remaining feature of the adversarial model is that the hard distribution is chosen per learner. This is not a modeling preference but a necessity at the standard sample size (Proposition E.2): against any *single*, publicly known distribution, every ε-learner can be immunized on it – and on a whole neighborhood of it, not a knife-edge – by a hybrid that outputs a fixed vector whenever a goodness-of-fit test passes. Distribution-free accuracy is exactly the property that forces the learner to encode its sample in $\hat{\theta}$; hardness must therefore be exhibited by *some* distribution, chosen per learner.

Proposition E.2 (Immunizing hybrid)**.** *Let $\mathcal{D}$ be any fixed, publicly known distribution over $\mathcal{Z} = \{\pm 1/\sqrt{d}\}^d$ (so $\|z\| = 1$ for all z), with mean $\nu = \mathbb{E}_{\mathcal{D}}[z]$ and population minimizer $\theta^*(\mathcal{D})$ of the linear loss ($\theta^*(\mathcal{D}) = \nu/\|\nu\|$ if $\nu \neq 0$, and any fixed $\theta_0 \in \Theta$ otherwise). Fix $\delta \in (0,1)$, let $c = 1 + \sqrt{2\log(1/\delta)}$, and let $n \geq 9c^2/\varepsilon^2$. Given any ε-learner $\mathcal{A}$, define the hybrid*

$$\mathcal{A}'(S) \;=\; \begin{cases} \theta^*(\mathcal{D}) & \text{if } \|\bar{z} - \nu\| \leq 2c/\sqrt{n}, \\ \mathcal{A}(S) & \text{otherwise,} \end{cases} \qquad \bar{z} = \frac{1}{n}\sum_{i=1}^{n} z_i.$$

Then: (a) *on every distribution over $\mathcal{Z}$, $\mathcal{A}'$ has excess risk at most 4ε with probability at least $1-2\delta$ – an ε-learner after rescaling ε and δ by constants; and* (b) *on every distribution $\mathcal{D}'$ over $\mathcal{Z}$ whose mean ν' satisfies $\|\nu' - \nu\| \leq c/\sqrt{n}$ – in particular on $\mathcal{D}$ itself – with probability at least $1-\delta$, $\mathcal{A}'$ outputs the fixed vector $\theta^*(\mathcal{D})$, so its output carries no information about the training set beyond the single bit that the test passed, and no reconstruction from it is possible.*

Proof. Excess risk of the fixed vector, exactly. For a distribution $\mathcal{D}'$ over $\mathcal{Z}$ with mean ν', the population risk of the linear loss is $F_{\mathcal{D}'}(\theta) = -\langle \theta, \nu' \rangle$, minimized over Θ at value $-\|\nu'\|$. Writing $\Delta = \|\nu' - \nu\|$, for $\nu \neq 0$,

$$\mathrm{exc}\big(\theta^*(\mathcal{D}); \mathcal{D}'\big) \;=\; \|\nu'\| - \Big\langle \frac{\nu}{\|\nu\|}, \nu' \Big\rangle \;=\; \frac{\Delta^2 - (\|\nu'\| - \|\nu\|)^2}{2\|\nu\|} \;\leq\; \min\Big\{2\|\nu'\|,\; \frac{\Delta^2}{2\|\nu\|}\Big\},$$

and this holds at *every* gap Δ, not only inside the test radius. The second branch is second order in the gap, giving the $\varepsilon/2$ bound below when $\|\nu\| \geq \varepsilon$; the first branch covers small $\|\nu\|$, and also the degenerate case $\nu = 0$, where the excess risk of any fixed θ_0 is trivially at most $2\|\nu'\|$. So the argument below applies to every fixed $\mathcal{D}$.

Tail for the test statistic. Every point of $\mathcal{Z}$ has unit norm, so for any distribution $\mathcal{D}'$ with mean ν', the map $S \mapsto \|\bar{z} - \nu'\|$ has bounded differences $2/n$ and expectation at most $\sqrt{\mathbb{E}\|\bar{z} - \nu'\|^2} \leq 1/\sqrt{n}$. McDiarmid's inequality gives, for every $s \geq 0$ and every distribution on $\mathcal{Z}$,

$$\mathbb{P}\big[\|\bar{z} - \nu'\| \geq (1+s)/\sqrt{n}\big] \leq e^{-s^2/2},$$

so with $c = 1 + \sqrt{2\log(1/\delta)}$: $\mathbb{P}[\|\bar{z} - \nu'\| \geq c/\sqrt{n}] \leq \delta$. The hypothesis $n \geq 9c^2/\varepsilon^2$ gives $3c/\sqrt{n} \leq \varepsilon$.

(a) Accuracy. Let the distribution at hand have mean ν' and gap Δ. If $\Delta > 3c/\sqrt{n}$: passing the test forces $\|\bar{z} - \nu'\| \geq \Delta - 2c/\sqrt{n} > c/\sqrt{n}$, an event of probability at most δ; so with probability $\geq 1-\delta$ the test fails, $\mathcal{A}$ runs, and $\mathcal{A}$'s own (ε, δ)-guarantee applies – total failure probability at most 2δ. If $\Delta \leq 3c/\sqrt{n} \leq \varepsilon$: whenever the test passes, the display above bounds the excess risk *deterministically* – by $\Delta^2/(2\|\nu\|) \leq \varepsilon^2/(2\varepsilon) = \varepsilon/2$ if $\|\nu\| \geq \varepsilon$, and by $2\|\nu'\| \leq 2(\|\nu\| + \Delta) < 4\varepsilon$ if $\|\nu\| < \varepsilon$ – while whenever it fails, $\mathcal{A}$'s guarantee applies. In all cases the excess risk is at most 4ε with probability at least $1-2\delta$.

(b) Immunity on a neighborhood. Let $\mathcal{D}'$ have mean ν' with $\Delta = \|\nu' - \nu\| \leq c/\sqrt{n}$. By the triangle inequality, $\|\bar{z} - \nu\| \leq \|\bar{z} - \nu'\| + c/\sqrt{n}$, so the test passes whenever $\|\bar{z} - \nu'\| \leq c/\sqrt{n}$, an event of probability at least $1-\delta$ by the tail bound. On it the output is $\theta^*(\mathcal{D})$, a vector fixed before the sample was drawn; by data processing, nothing about S is recoverable from it beyond the single bit that the test passed. The case $\mathcal{D}' = \mathcal{D}$ is $\Delta = 0$. □

The hypothesis $n \geq 9c^2/\varepsilon^2 = \Omega(\log(1/\delta)/\varepsilon^2)$ is the standard sample size for excess risk ε at confidence δ, and it lies above the window $n \leq 1/(13824\,\varepsilon^2)$ of Theorem 3.2. The proposition is a statement about the adversarial model (at that sample size, no theorem with a single fixed hard distribution is possible), not about that window. Two observations mitigate the per-learner choice, and both are true of our construction:

1. *The choice is minimal and almost learner-oblivious.* The family, the prior $\mu \sim \mathrm{Uniform}([-12\varepsilon/\sqrt{d}, 12\varepsilon/\sqrt{d}]^d)$, and the partition-sampling and bit-fitting rules are fixed with no reference to the learner; only the realized μ (and through it the pin values) depends on it, extracted by an averaging argument over that fixed prior (Lemma B.2), as in Attias et al. (2024).
2. *Equivalently, a statement about a fixed class.* Theorem 3.2 reads as: no ε-learner resists reconstruction on every block-uniform distribution in this family. Under this reading the adversary "chooses" nothing.

E.3 Decodable Category Families

The block-uniform construction of Section 3 is one member of a broader class of families to which the same argument applies.

Remark E.3 (Decodable category families)**.** *Besides the correlation bound (Lemma B.2) and $N = n^2/\delta$ disjoint categories (Definition 3.1; the collision step of Theorem 2.4), the exponential-domain argument uses four properties of the constructed family:*

(i) polynomially many categories*;*

(ii) per-category decodability*: from the learner's output $\hat{\theta}$, a single candidate per category is computable in polynomial time and, with probability $1 - O(\delta)$, equals every training point of that category that clears the test threshold (here: block-constancy plus sign decoding, Algorithm 2 and Lemma B.4);*

(iii) a sub-Gaussian category tail*: for the relevant test vectors y, the statistic $\langle y, z - \mathbb{E}[z \mid \text{category}]\rangle$ has a sub-Gaussian tail (here Lemma 3.3, replacing the product-measure tail bound of Attias et al. (2024), proven from the block-sign representation alone);*

(iv) mean alignment*: the category means are aligned with the target parameter, in two senses – $\max_l |\langle y, \mathbb{E}[z \mid l] - \mu\rangle|$ lies below the test threshold for the relevant test vectors (in our construction via the center-shift bound, Lemma C.1, as established in the proof of Corollary 3.4), and $\mathbb{E}_\Delta[z]$ is close enough to μ that the accuracy-to-correlation transfer costs less than the detection margin (in our construction, the role of the pinned coordinates and the bit-fitting rule).*

Any "decodable category family" satisfying (i)–(iv) supports the proof of Theorem 3.2; uniformity within categories is a convenience, not a necessity – biased within-category distributions with bounded sub-Gaussian norm satisfy (iii) with adjusted constants. We state this as the interface for future instantiations.

E.4 Approximate Reconstruction for Continuous Distributions

Under a continuous data distribution, outputting a training point exactly is a measure-zero event, so the reconstruction goal itself must change: the standard notion is *approximate* reconstruction – output z' with $\|z' - z\| \leq \rho$ for some training point z (Balle et al., 2022). The reduction template extends to this goal. Run Algorithm 1 over a ρ-net of the support in place of the sampled candidate set, and rely on the test having a *margin*, so that a training point is still detected through its nearest net point. The false-positive union bound then runs over the net size C rather than $|\Sigma|$: whenever the test's false-positive rate is $o(\delta/C)$, the argument of Theorem 2.4 recovers $\Omega(n \cdot q(n))$ training points up to error ρ – for instance, for bounded-support distributions in fixed dimension, where a polynomial-size net exists. Our SCO test does have such a margin: the statistic $x \mapsto \langle g(\varepsilon, \mu) \odot (x - \mu), \hat{\theta}\rangle$ is Lipschitz in x with constant $\|g\|_\infty \|\hat{\theta}\| \leq 288\varepsilon^2$, while soundness holds already at threshold $\varepsilon/(2n)$ when $n = \Theta(1/\varepsilon^2)$ (Corollary 3.4); testing net points at threshold $\varepsilon/(2n)$ instead of ε/n, a training point with statistic at least ε/n is detected through its nearest net point whenever $288\varepsilon^2\rho \leq \varepsilon/(2n)$, i.e. $\rho \leq 1/(576\,\varepsilon n)$. This is the same slack that Appendix E.1 spends on estimation, so combining the two splits it.

We state the caveat explicitly: *no end-to-end continuous theorem is claimed.* A ρ-net of polynomial size requires essentially fixed dimension, while our test's margin is proven in the high-dimensional regime of Theorem 3.2 ($d = \Omega(n^8 \log^4(n/\delta))$), so the two ingredients do not currently meet: an end-to-end continuous theorem needs a

membership test at the required rates whose margin lives over a polynomially sized net. What the mechanism does show is that continuity is no obstacle to the reduction template itself; we regard closing this gap as a concrete open direction, and the exponential-family route below as the most promising path toward it.

E.5 Beyond Product Bernoulli: Exponential Families

The sole component of our chain tied to the product Bernoulli family is the correlation bound (Lemma B.2, following Lemma 6.3 of Attias et al., 2024), which converts ε-accuracy into per-sample signal and rests on a fingerprinting lemma for that family (Bun et al., 2014; Steinke and Ullman, 2017). The fingerprinting method is no longer confined there: Kamath et al. (2022) prove a fingerprinting lemma for arbitrary exponential families $\mathcal{E}(T, h)$ (their Lemma 3.2), with the natural parameter drawn uniformly from a box and the correlation measured against the sufficient statistics $T(X_i)$; no privacy assumption enters that step, no coordinate-independence of the samples is required, and the binary-product and Gaussian lemmas are recovered as special cases.

We therefore state as our main open problem: *establish a membership inference impossibility at the rates the reduction consumes – recall at least $q(n)$ and false-positive rate $O(\delta^2/n^2)$ under the hard distribution of Definition 2.2 (Definition 2.3) – for general exponential families.* Three steps appear to be needed: (1) re-parameterize the SCO instance so that excess risk controls the estimation error of the natural parameter η (a natural candidate is the negative log-likelihood $f(\theta, x) = Z(\theta) - \langle \theta, T(x) \rangle$, convex because the log-partition function Z is); (2) read the per-sample signal off $T(x)$ rather than x – the two coincide in our instance, where $T(x) = x$, which is why the current proof need not distinguish them – making the reconstruction target a sample through its sufficient statistics; and (3) supply a concentration bound for the new family in place of Lemma 3.3. Since the reduction (Theorem 2.4) is distribution-agnostic, any such result upgrades to reconstruction with no further work, covering continuous families such as Gaussians and removing the Bernoulli assumption rather than re-proving it family by family.